

Business and Manufacturing

In the fast-paced world of business and manufacturing, competitive advantage can mean the difference between a successful endeavor and one that is surpassed by the competition. Being the first to introduce a product to the market may establish a company as a leader. A company's ability to harness the power of new technology can enable it to exploit its own competitive edge. Businesses also succeed by building and controlling market share. The novel medium of VR can often be exploited to generate interest in a company or product.

Early adoption and exploitation can give a competitive advantage. Three primary ways virtual reality has been used to increase competitiveness include product development, business education, and marketing. The underlying goal of exploiting VR in business and manufacturing has been, of course, to enhance profitability and/or productivity. Many have found that VR offers a means to improve productivity, increase safety, speed-up design cycles, raise product awareness, and other strategic advantages.

By using VR to increase productivity, an additional benefit can be gained through a variety of subtle marketing strategies. Companies using VR in new ways are enjoying considerable press coverage, which also enhances their public image.

The steps toward utilizing VR in business have been occurring since the advent of computer-based drafting and design systems. Companies that adopted such tools found they were able to make decisions and adapt to changing conditions quickly. Virtual reality is an extension to the evolution of the physical world into the digital realm. Often, physical models and prototypes will still be essential, but with VR it is possible to delay and minimize the creation of such models, thereby saving costs.

In its early inception, virtual reality was found mostly in research and development (R&D) departments. R&D departments are tasked with finding new

technologies to help their company make more money. The pattern has been that after initial seed money is spent creating example VR applications, those researchers are required to seek out those within their company who are interested in using the applications. If there is not enough interest to justify further research on an application or technology, research typically ceases.

However, it may be renewed when other advances indicate an increase in its potential advantage.

3.1 AREAS OF APPLICATION

There are many ways in which VR has been applied in business. Some have been explored more than others. This chapter describes several applications of VR in the area of product development as well as examples of usage for training and marketing. Research has also begun on the use of VR for analyzing production flow, especially for factory floor machine layouts.

3.1.1 Product development

Perhaps the most widely explored usage of VR by the business world has been in product development. This includes the use of VR for *virtual prototyping*, which is the use of a computer to simulate a design of an object or system. While virtual prototyping does not require the use of VR, the two are often associated.¹ Examples of virtual prototyping that have successfully incorporated VR addressed in this chapter include efforts by Caterpillar Inc. and General Motors Corp. Another interesting example includes experimental studies performed by DaimlerChrysler on the perceptions of customers to possible car interior designs. VR has the potential to be used as a tool in which a product is designed. This is demonstrated in the collaborative work between Nalco Fuel Tech and Argonne National Laboratory, where optimal placement of fuel injectors within a boiler can be determined via an immersive, interactive tool (described in Section 3.4). Work has also been done by Lockheed/Martin to create a ship-designing tool that allows immersed participants to manipulate designs, with consequences such as pipe relocation performed automatically.

VR has also been tested as a means for improving productivity during the manufacturing process. In particular, The Boeing Company created an experimental augmented reality system that shows a human assembler of aircraft wire-bundles where to place each wire, a step at a time.

3.1.2 Business education

The use of VR as a training tool is spread across many fields. One notable example in the area of manufacturing is the training of assembly line procedures at Motorola University.

¹ An example of virtual prototyping that does not use VR is electronic circuit design, where a simulation of the circuit allows the results to be measured, and compared with expectations.

Motorola found that VR benefits the learning process in three ways: 1) the virtual assembly line is much cheaper to create and duplicate than a physical practice line; 2) the computer version is easier to transport to other training centers; plus 3), the workers being trained are more highly motivated to work with the virtual assembly line than previous training methodologies.

3.1.3 Marketing

VR is sometimes used as part of a marketing strategy to demonstrate that a particular company or product is high tech, forward looking, or just plain cool. In this case, getting people to remember and talk about the product is the goal. As a new, exciting, and often talked about technology, virtual reality has been used several times as part of advertising campaigns. Television advertisements include those by Honda, Starburst, and Bubble Yum.

Virtual Reality gives the opportunity to advertisers of providing advertising that people want to participate in. One way in which this has been done was through the development of a specific VR game that relates to the company or product. This game was then taken to several venues (stores, trade shows, etc.), where people literally stood in line to experience the advertisement. Since lines often attract attention to themselves, the effect often snowballs, bringing more people to see the message. In this chapter, we describe the usage of VR for marketing Cutty Sark scotch. The Fraunhofer Institute for Computer Graphics (IGD) has also done a project for a Swiss bank to help draw attention to a new bank card they were introducing.

The VR application (game) itself might be a good piece of advertising. However, because of the excitement, there are usually several people queued up to take their turn. This presents another opportunity to communicate with potential customers. While in line, people can be presented product information in more traditional ways as well as provide information back to the company by filling out marketing questionnaires. Most people do not object to providing this form of "payment" to try the experience. Examples of this include Virtuality for the Ford Galaxy (Chapter 9) and Matsushita for Kitchen and House products.

[Du Pont, P. (1995). "VR for thermal visualization: Analyzing air and temperature flow," VR world, May/June, 58–59.

hotwired.lycos.com/collections/virtual_communities/2.07_kitchen.html

Often, the use of VR in other areas of business and manufacturing has the secondary result of bringing attention to the company. This is generally intentional, and part of the company's marketing effort. It is likely that as VR becomes a less novel technology, this particular use of VR as marketing tool will wane.

3.2 OTHER AND FUTURE USAGE OF VR IN BUSINESS AND MANUFACTURING

As expertise in implementing VR applications grows, and the technology becomes more economical, it will be applied to many other areas and types of business.

Some research that is already underway includes the usage of VR as a tool for studying production flow, especially for factory floor machine layouts. Research has also been done on maintenance analysis of product designs, taking advantage of the LBE game *BattleTech* and using a simplified version of the experience as part of a team-building effort.

Another area in which VR has been used extensively includes collaborative work environments, allowing workers to more easily work with others who might be at a remote site. An example of this is the *Virtual Director* application described in Chapter 9.

Other potential areas for applying VR to business and manufacturing include the visualization of business information and farming data. Business visualization is currently a growing area of research among visualization experts, and has expanded into the medium of VR. Similar techniques will also be useful for people who need to access large archives of business data. A spatial organization of information can be helpful in being able to sort through such data. VR can provide a means of allowing a person to spatially experience a database. Spatial data in VR can be achieved by extending typical GIS representations into 3D, and also by developing completely new representations. Farming has become increasingly high tech. Many farms now use the global positioning systems (GPS), combined with soil analysis techniques, to apply specific fertilizers/pesticides for certain locations in the field. This and other information can be transferred to a VR visualization tool for analysis and/or telepresence applications for farmers.

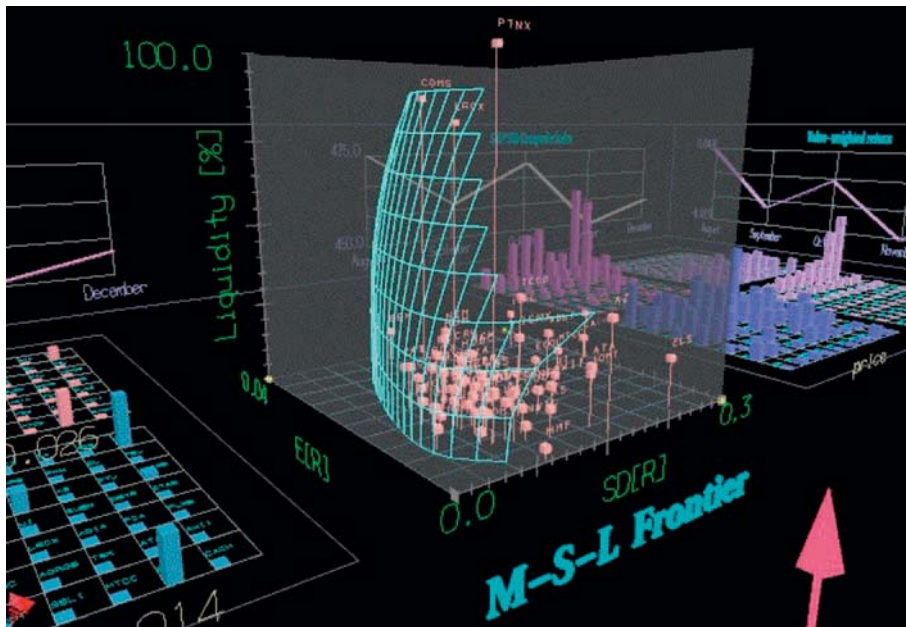
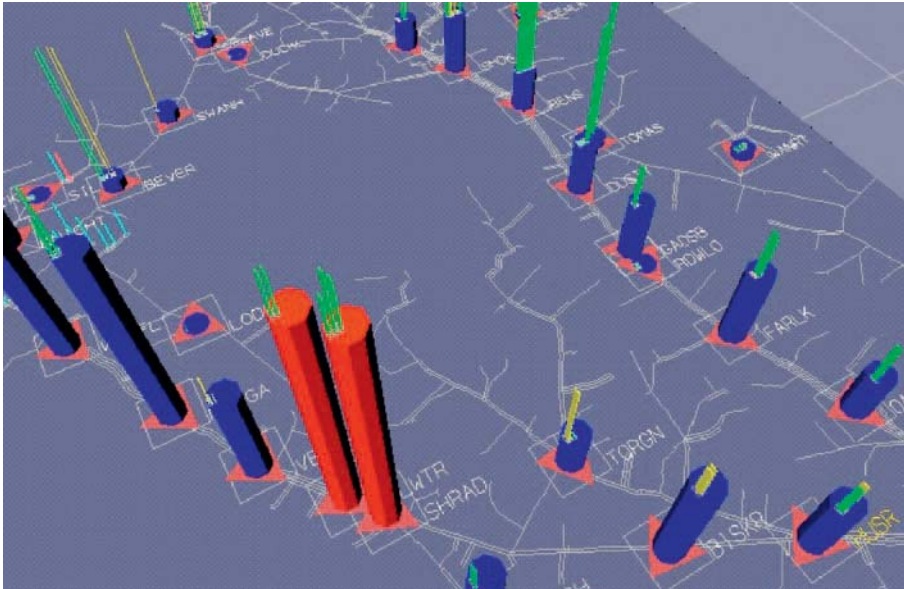


FIGURE 3-1

Virtual reality has allowed new visualizations of business information. These images from a portfolio analysis tool (above) and an oil pipeline monitor (the next page) show examples of such representations.

Photos courtesy of Visual Decisions, Inc.

**FIGURE 3-1***(Continued)*

CASE STUDY 3.1: USING AUGMENTED REALITY FOR ADVERTISING

The Wellington Zoo

Executive Summary

Traditional print advertising suffers from several deficiencies that are difficult to overcome. Among them are that imagery is only visible as flat, two-dimensional pictures, and that they are required to be static. Until now, The Human Interface Technology Laboratory New Zealand (HIT Lab NZ) at the University of Canterbury in Christchurch, New Zealand, has successfully demonstrated that augmented reality techniques can allow advertisers to purchase a relatively small, two-dimensional, black-and-white advertisement in a newspaper, or magazine, and yet convey colorful, dynamic, three-dimensional virtual images. The way it works is that a customer sends a text message to obtain a small computer program that runs on their cellular telephone. The advertiser places a specially designed, black-and-white marker on the print ad. When the customer points the camera of their cellular telephone at the marker,

they can see and interact with the full-color, three-dimensional virtual world overlaid on the real print advertisement.

**FIGURE 3-2**

Potential zoo visitors are able to see a 3D rendering of a cheetah by pointing their cellular telephone at a newspaper ad. By moving the telephone around, they can see the animal from different perspectives.

The Wellington Zoo, in Wellington, New Zealand, is the first organization to exploit the advertising potential of the augmented reality (AR) medium. They wanted to create a dynamic print advertisement for their “Close Encounters” exhibit. Working with the Saatchi & Saatchi NZ advertising agency and the HyperFactory² mobile marketing provider, and powered by the AR technology developed at the HIT Lab NZ, they placed a print advertisement in one of New Zealand’s largest newspapers, with a circulation of 750,000 people. Readers downloaded the software to their cellular telephone by text message, and then were able to interact with three-dimensional virtual views of zoo animals by pointing their telephone at the marker in the advertisement. They could see the virtual animals from all angles by moving their telephones around the image. The advertising was effective, with the zoo realizing a 32% growth in visitors and the campaign winning numerous international advertising awards.

Application Description

Representation of the Virtual World

One of the goals of this application was to allow potential zoo viewers to see some of the animals from the “Close Encounters” exhibit up close and from all perspectives. As part of this advertising campaign, the developers created



FIGURE 3-3

This person is reading a newspaper and pointing a cellular telephone camera at a marker in the paper. The cheetah is visible from different perspectives by moving the telephone in relation to the newspaper.

²<http://www.thehyperfactory.com/>

three 3D computer graphic modeled animals: a cheetah, a giraffe, and a bear.

The animals were modeled with an eye toward realism. That is, the cheetah model looked like a cheetah. Each animal was set on a grassy expanse to show it more or less in an appropriate environment.

VR DISCOVERY

Augmented reality advertising works. The zoo experienced a 32% increase in visitors. The techniques shown by this application can also be used on other print media, including books, posters, or even the side of a building.

DESIGN CONSTRAINT

Because the virtual world would be displayed on the tiny screen of a mobile phone, the developers primarily exhibited one animal at a time. The background was straightforward and showed the animals in the open, to allow a good view of them, and to avoid cluttering the display with background objects.

The application was only able to run on certain models of Nokia mobile phones (the N-series Nokia smart phones). So when the user sent a text message to the number shown in the print advertisement, back-end server software would first check to see if they had the one of the supported phones. If so, they were sent a 200 kB application that enabled them to view the AR animal models. If their phone was not supported, then they were sent a static image showing them what the application would have looked like running on their phone if they had a suitable phone model.

Although this application did not make use of audio to keep the download file size small, there is no technical reason why the telephone could not be used to play animal sounds, or an audio advertising message. In the normal mode of operation, the end user typically holds their phone



FIGURE 3-4

The Wellington Zoo print advertisement as it appeared in the paper. The black square is used by the AR application to track the camera phone position and overlay a virtual zoo animal on the paper page.

in front of them as they pan around the virtual world. Hence, to effectively utilize audio, the phone would need to be placed in speakerphone mode.

Interaction with the Virtual World

The typical way an end user would interact with this application is to point the camera of their phone toward the real printed marker (known as a fiducial marker) and view the virtual image on the telephone display. On the telephone display the virtual graphics are combined with live video from the real world, so the 3D model appears overlaid on the real print ad.

In order to see the virtual world from a different perspective, the participant has two primary options. The most obvious way to navigate about the virtual world is to physically move the cellular telephone around the marker. Thus, you can pan around the space just by moving the telephone (keeping the camera pointed at the marker). The other primary way to navigate about the space is to keep the phone in one position and move the newspaper, or other print media, around. This method might be more convenient if one were, say, sitting at a table reading the newspaper. One could point their phone at the marker on the paper, and then maneuver the paper around on the table to see the virtual world from different perspectives.

In this simple AR application the user did not need to use the keypad to interact with the virtual animal, and only one animal was shown. During the campaign three different print ads were presented in the newspaper and so users needed to download three applications to see all of the different animals. This particular application is not a collaborative experience, but mobile AR technology is conducive to collaborative applications.

Application Implementation

This application was created using the ARToolkit computer vision tracking library. ARToolkit³ is an open-source library for building augmented reality applications that was initially created at the Human Interface Technology Laboratory at the University of Washington in the United States. Anders Henrysson⁴ ported this software to the Symbian operating system for mobile phones, and the HIT Lab NZ used this port to create the mobile AR advertising

³<http://artoolkit.sourceforge.net/>.

⁴Henrysson, A., and Ollila, M. *UMAR—Ubiquitous Mobile Augmented Reality* In Proc. Third International Conference on Mobile and Ubiquitous Multimedia (MUM2004) pp 41–45. College Park, Maryland. October 27–29, 2004.

**FIGURE 3-5**

HIT Lab NZ director Mark Billinghurst uses a handheld display to view a "Magic Book" in collaboration with a colleague. Each person sees the virtual world (and the real world) from his or her own perspective.

application. The toolkit provides services such as the computer vision required for the optically based tracking, thus providing a real-time 6 Degree of Freedom marker tracking system. The toolkit enables the overlay of the 3D virtual objects on the fiducial markers. The 3D virtual objects were created by using the Maya modeling software and the initial complex models simplified down to a few hundred polygons in size. The final application delivered to the mobile phone used the OpenGL ES mobile graphics library to render the 3D virtual models.

The VR System

The Wellington Zoo advertising application was developed primarily for use on mobile telephones. However, augmented reality technology can be supported on other hardware platforms. The most basic hardware requirements for augmented reality are a camera, a computer, and a display. Thus, there are a number of other physical scenarios in which augmented reality can be done. One common example is the use of a laptop computer with a web camera. In this case the end user can point their web camera at a fiducial marker and see the 3D virtual world appearing on their screen. At the higher end of cost and complexity are specialized head-mounted or handheld displays that have

**FIGURE 3-6**

Augmented reality can be experienced stereoscopically by using specialized see-through displays with an attached camera. This provides the most "authentic" AR experience.

cameras integrated into them. By using a head-mounted display, a user can see the virtual world from exactly their correct perspective, and also see the images rendered stereoscopically.

Conclusion

The Wellington Zoo application has been a smashing success on many fronts. Although mobile phone-based augmented reality had previously been demonstrated on mobile phones, this was the first time the technology had been used in a real-world advertising campaign. The advertising campaign led to a 32% increase in visitors to the zoo's "Close Encounters" exhibit. The advertising has earned numerous awards and honors. More importantly, this application clearly indicates that augmented reality has turned the corner from being a laboratory novelty and has been used successfully in the commercial world. The HIT Lab NZ's accomplishment with AR technology has been recognized, and they have since been engaged to develop other advertisements and games on cellular telephones that exploit augmented reality.

**FIGURE 3-7**

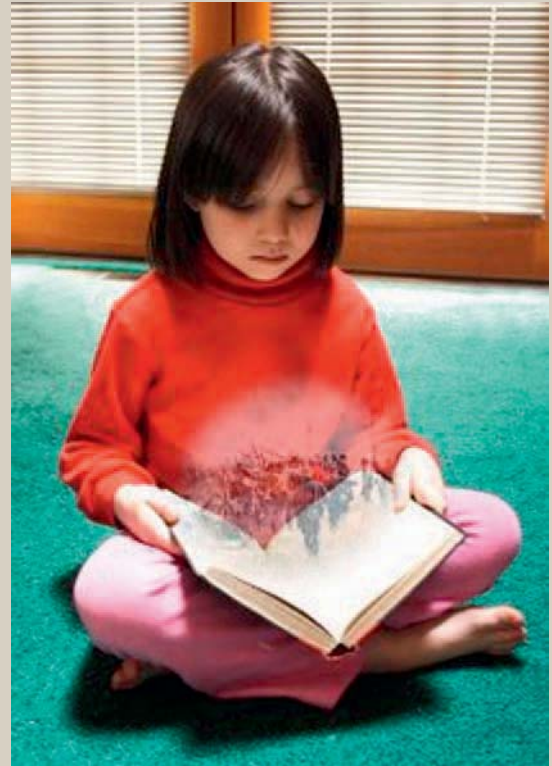
This is an example of an augmented reality fiducial marker. The ARToolKit library provides tools to create, and to read such markers. Notice the black frame, and the nonsymmetry of the image.

**FIGURE 3-8**

HIT Lab NZ's director, Dr. Mark Billinghamurst, reads a "Magic Book" in which he not only sees the text and images of the book but also sees a three-dimensional model of a sailboat. If he wishes to see the other side of the ship, he can turn the book around. The ship can also be animated to see the sails flapping in the breeze.

Related Applications and Follow-on Work

The Wellington Zoo advertising application follows on a long history of augmented reality applications developed by HIT Lab NZ, and their sister center, the HITLab at the University of Washington in the United States.

**FIGURE 3-9**

The Institute for Computing in Humanities, Arts, and Social Science at the University of Illinois at Urbana-Champaign is exploring the use of "Magic Books" in education. Here is an artist's conception of a student reading a book about the Battle of Gettysburg. In the future, augmented reality will not require intrusive hardware.

Photo montage by Dave Bock

Perhaps the most widely recognized VR applications prior to this advertising campaign were the "Magic Books" that use augmented reality to allow readers to see 3D, animated augmentations of traditional physical books, in place, and registered spatially with the physical book.

I-CHASS, the Institute for Computing in Humanities, Arts, and Social Science at the University of Illinois at Urbana-Champaign, is exploring the use of augmented reality (also using the ARToolKit) to enhance books with animated, three-dimensional scenes for educational purposes.

Other laboratories, such as the augmented environments laboratory at the Georgia Institute of Technology, the Graz University of Technology, and others, continue to develop core augmented reality technologies and applications. The ICT Center at the Commonwealth Scientific and Industrial Research Organisation (CSIRO) in Australia has integrated Augmented Reality with haptic feedback devices.

Another example of AR in advertising was at the 2008 Auto Show in Los Angeles, California. Nissan used a brochure that was enhanced with augmented reality that could be viewed with a web camera. Indeed, a search of the world-wide web shows that AR advertising is catching on very quickly. An especially interesting example is the Smart Grid Augmented Reality Web-based system by General Electric.



FIGURE 3-10

Sony's Eye of Judgment game for the PlayStation 3 uses a computer monitor, a camera, and trading cards with AR markers on them to provide a novel and exciting game experience.

On a different but related note, augmented reality has already been commercially used in the gaming market. Sony released its *Eye of Judgment* game for the PlayStation 3 in October of 2007. The game provides a number of "trading cards" that contain a graphical image, plus an AR marker. Players place their cards on the table, and a camera on a stand next to the playing area identifies the markers and the game responds appropriately. Since its release, *Eye of Judgment* has sold more than 250,000 copies, making it the most widely distributed commercial AR application.

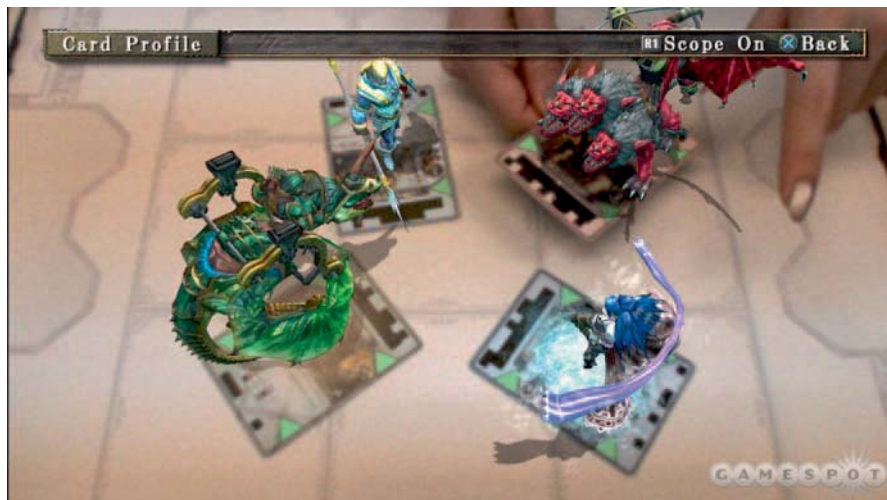


FIGURE 3-11

Sony's Eye of Judgment game allows players to move their cards near their opponents' cards to engage in battle.

CASE STUDY 3.2: CATERPILLAR VIRTUAL PROTOTYPING SYSTEM

Executive Summary

The Caterpillar Virtual Prototyping System (VPS) was created to allow design engineers to test new machine designs prior to building physical models. The preliminary usage of the system was for assessing operator visibility and allowing engineers to make corrective modifications very early in the design process. Later, the system was expanded to include the capability to evaluate a variety of different aspects including machine performance. Development of the system focused on enabling engineers located at geographically distant sites to collaborate on projects by sharing the same virtual world.

The initial VPS used head-mounted display technology to present the view of the virtual landscape. That technology was replaced by projection-based displays such as *CAVE*[™], *ImmersaDesk*[™], and *Responsive WorkBench*[™]. A prominent part of the system includes a physical platform with

an actual machine seat, steering wheel, pedals, and other control elements that are used to control the prototype systems in a near-normal operational setting. A dynamic model simulates the operation of the virtual machine. Additionally, a very simplified program gives the illusion of soil dynamics. The soil model was integrated to create typical scenarios for operating simulated earth-moving machines. The VPS has been involved in actual product design, including the Caterpillar® 914G Wheel Loader.

Introduction

Caterpillar, Inc. has been designing and manufacturing earth-moving machinery since the 1920s. Although machines have changed immensely over this period, one thing has remained constant. To be competitive one must constantly strive to stay one step ahead of the competition. Studying what the customer needs and providing it sooner helps to accomplish this goal.



FIGURE 3-12

Caterpillar, Inc. was an early adopter of the use of virtual reality systems for virtual prototyping. Shown here is Kem Ahlers navigating through a representation of the Caterpillar proving grounds.

To remain competitive requires Caterpillar to be leaders in the use and development of new technology as an aid in the design and manufacturing process. Caterpillar has been using computer simulation and modeling in the development of their vehicles since the 1960s. More recently, they began using scientific visualization and computer graphics to aid them in the design process.

Although batch simulation and visualization have proven useful, they do not support the level of interactivity required to include the “human in the loop” in the simulation—that is, to enable a human operator to “drive” the machine in real time with the simulation taking into account the human’s actions. Designs that rate poorly based on static tests and simulations may actually be reasonable. For example, in a static test, an operator’s view might be impeded by some obstruction. By integrating the operator’s head position into the simulation, the operator may be able to adjust their seat and simply look around the obstruction.

The *Caterpillar VPS* went through several phases. Throughout the project’s development, a physical machine cab where an operator sits and uses actual tractor controls was the primary means for interacting with a computer simulation of the machine. In the first phase, head-mounted display and single-screen projection systems were used as the display mechanism to evaluate the visibility of a virtual backhoe loader. In subsequent phases of the application, a *CAVE* projection-VR system has been used as the primary display device, and has been used to examine the design of many other machines, including a variety of front-end wheel loaders.

The *Caterpillar VPS* project encapsulates three overall goals:

- Shorten the time and cost in new product introduction
- Consider an order of magnitude more design alternatives
- Improve “early hour” discovery of problems

Achieving these goals has helped Caterpillar to be competitive in the global marketplace. Strengthening competitiveness is achieved by using simulation as a tool to understand how a product will behave for the customer. By developing

virtual prototypes of a product, Caterpillar has uncovered such knowledge early in the design process, allowing changes to be made while changes are still easy and inexpensive to make. As a product’s design moves from concept phase to design phase, changes to the design increase in cost by tenfold over what they would have cost in the concept phase. When proceeding from design to production, the cost of a change increases by an order of magnitude. Once the product is in the customer’s hands, the cost of fixing a problem is even greater. Thus, it is extremely advantageous to find any problems as early as possible in the process to remedy them inexpensively.

In addition to finding “fatal” errors in a design that must be corrected, virtual prototyping enables designers to make more iterations. Each iteration can improve the product in many small ways, ultimately resulting in a better product at a lower cost. Also, it is easier to compare different design alternatives. Virtual prototyping enables up to 10 times as many design alternatives to be tried than by conventional methods.

Prior to using the VR virtual prototyping system, designs were modeled using physical prototypes. These prototypes were very expensive and took considerable time to build. Physical prototypes are still the norm for most machine development. The VPS has allowed Caterpillar to create fewer physical prototypes.

Caterpillar has used computer simulation of products for many years, as well as desktop computer-aided design (CAD). In the early 1990s, Caterpillar Inc. enlisted the aid of the NCSA Visualization Group to collaborate with them to explore the possibilities of using computer graphics and animation to aid them in prototyping vehicles. The earliest efforts in this joint research explored what could be learned about operator visibility problems in a particular machine. The goal was to learn of problems *before* building physical model of the vehicle. NCSA animator Mark Bajuk developed frame-animation techniques to aid in assessing operator visibility in new tractor designs that existed only within their computer CAD and simulation models. Caterpillar already had the CAD databases for their designs, as well as real-time simulations of the vehicle dynamics. The next step was to incorporate real-time computer graphics, user tracking, and operator interaction.

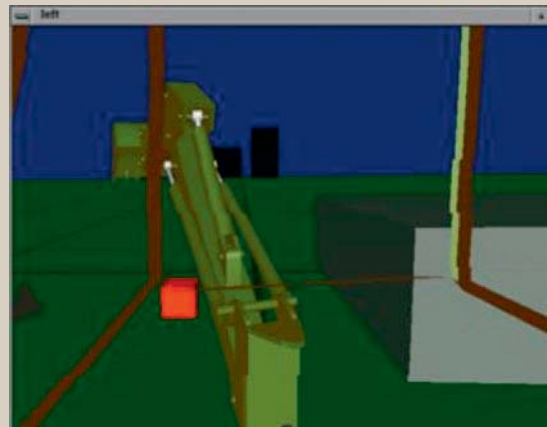
**FIGURE 3-13**

The image on top is an example of an early visualization to assess operator visibility. This image is a frame from an animated movie created by NCSA. The image on the bottom shows the view from an early example of the VPS with the operator controlling the system interaction.

Application Description

The virtual prototyping system is meant to serve many purposes. Some of the uses Caterpillar envisions for the use of the virtual reality system include:

- Ergonomic studies
- Machine performance in the work cycle
- Serviceability audits
- Manufacturing and assembly
- Marketing
- Training

**FIGURE 3-14**

The original Caterpillar VPS made use of a head-mounted display (above). The image below shows a view of the original, simple virtual world.

Since the original project was to assess operator ergonomics through the complete work-cycle of a virtual backhoe, a physical platform that emulated the controls of an actual backhoe was built to allow the operator to control the virtual backhoe. To view the world, the operator was placed in a head-mounted display.

In this system, the user was able to see (from the first-person point of view using the HMD) and control the arm and bucket of a backhoe loader. The tractor was stationary (as in real life). There were simple cubes in the virtual scene that a skilled operator was able to lift with and drop from the bucket. The operator in the head-mounted display was

unable to see their own hands, so an assistant was required to help them find the control levers when first “suited up.” In this installation visibility analysis with an operator free to move their head was possible. They were also able to get out of the cab, and walk around the tractor and see the tractor from the appropriate points of view, although to help rendering performance only the most important parts of the tractor were modeled.

Building on the success of the original backhoe system, the *Caterpillar VPS* was extended to simulate a wheel loader, and a control platform was built to match. The front end loader was much more complex, adding driving controls such as steering wheel, gear shift lever, accelerator, clutch, and brake to the levers required for controlling the bucket and lift arms. These functions not only were added to the physical control mechanism (the platform) but also to the world dynamics simulation. The visual scene of the virtual world was also raised in complexity by changing the

small generic yard previously used for practice into a more complex area with a variety of visual elements including buildings, piles of rocks, and other entities. The loader and a dump truck were also rendered more realistically.

Representation of the Virtual World

The virtual world in the later editions of *Caterpillar VPS* consisted of models and simulations of a variety of real locations where new tractors are tested under a variety of conditions: a mining pit, a factory, a sanitary landfill, and a highway construction site.

Virtual world physics are calculated by a very simplified computational model of the dynamics and kinematics of the machine being prototyped—including the powertrain, hydraulics, wheels and tires, steering system, and suspension. While models for engines are available, the models used in *VPS* are simple mapping functions from gear selection, throttle and breaking to a speed. There is also a very



FIGURE 3-15

As part of the *Caterpillar VPS*, a simple model of the Caterpillar proving grounds in Peoria, Illinois, was used to enhance the operator's experience.

Image courtesy of Caterpillar, Inc.

simplified soil model that portrays frictional soils such as gravel, sand, or coal. Complex, highly realistic soil models are difficult to compute in real time, so a simple, volume conserving model was implemented. The soil model allows the machine operator to test scooping and digging operations, on the delivery of soil to a truck, bin, or pile.

The *Caterpillar VPS* is an interesting application in that one of the goals is highly realistic representation. In this sense it is similar to a traditional flight simulator, with the notable difference that since it is a design tool, it is essential to be able to make changes to parts of the virtual world very readily. Caterpillar engineers are able to put a new design in the *CAVE*, rapidly utilizing the tools they have built for converting CAD data into a format suitable for the *VPS*. Since the goal is a realistic simulation, many decisions such as how the user will navigate in the system are very simple. The user should navigate in the same way they do in the real world; for example, operation involved pressing the gas pedal and rotating the steering wheel.

For a prototyping system it is also important for realism to extend to accurately representing the scale of the object as life size. This is important for doing studies that involve how the user will interact with the product. Accurately registering the size of the virtual object to the size of the object in the real world also allows testing of maintenance procedures (such as checking to see if a mechanic can reach an object or fit a tool into position to make an adjustment). By enabling designers to manipulate a ruler in the virtual space they can take measurements, but if the world is not to scale, then they may not trust those measurements.

Internally, the objects in this application are represented in a polygonal manner. These objects initially come from CAD engineering files. However, because these CAD files are very detailed and complex, the models must be simplified before being imported into the *VPS* to reduce the number of polygons required by each object.

The system designers believe that realism in interaction is of higher importance than attempting to achieve “photorealism” in images. This is in contrast with the approach taken by the General Motors VR design team, who consider photorealism to be more important than real-time performance for their application (see next section). Each group

made the decision based on the goals for their application. In the case of GM, the goal is to allow users to see what the product would look like visually, whereas Caterpillar is more interested in those aspects that require the “human in the loop,” such as how the machine performs and how well the operator can see from any position, or state of the machine.

The original computer graphics visibility animations were done outside of real time, in a batch mode. This yielded very high-quality imagery. For the VR system, real-time performance was required. Thus, the graphics were computed at much lower resolution, and with much lower scene detail.

The operator is able to see “out the window” views in any direction covered by the field-of-regard of the display device. The operator can also see a realistic representation of the interior of the cab, including a control panel overlaid with instrument status indicators. Some parts of the control panel are fixed texture maps, others are rendered in each frame to reflect the state of the simulation.

In the *Caterpillar VPS*, the machine is represented as visually accurate as possible within the constraint of real-time rendering. Representations of additional information such as stresses or other invisible aspects of the machine were not added to the visual rendering of the virtual world. The goals were more along the lines of seeing whether or not the bucket was visible throughout a work cycle, and with the operator’s head in various locations.

The designers feel that audio cues help to augment the sense of manipulating the machinery. This sense of presence is important to help operators be less conscious of the artificial environment, and to behave more naturally as they perform the task. If the system seems “odd,” operators may focus on that aspect of the virtual world instead of the task under study.

As with the visual representation, realism with the audio representation is the main goal. Both the old and new versions of *VPS* have made extensive use of sampled waveforms of recordings made from actual operating machines under real work conditions. Realistic examples of sample sounds include the engine, back-up indicator alarm, and pieces of the machine banging and clanking as they go through the paces of a typical work cycle.

**FIGURE 3-16**

Left: An operator seated on the platform can see appropriate “out the window” views from the virtual machine. Right: Bob Fenwick raises the bucket of a simulated loader in the CAVE. The instrument panel is visible and emulates its real-world counterpart.

**FIGURE 3-17**

The Caterpillar VPS attempts to render the environment as realistically as possible while maintaining a frame rate suitable for real-time interaction.

The physical platform the operator sits on and manipulates to control the machine is a prominent and key component of the *Caterpillar VPS*. The platform is equipped with an actual tractor seat, steering wheel, gas and brake pedals, gear shifter, and control levers. Each of these components provides haptic feedback to the operator that closely resembles that of an actual machine. The main difference is that rather than controlling hydraulics, these mechanisms provide electrical signals to provide feedback to the computational simulation.⁵ None of the controllers provide force feedback other than that innately provided by the spring loading of the control.

⁵In fact, “fly-by-wire” controls are pervasive in present-day systems. Control levers produce electrical signals that are interpreted and converted to inputs to the hydraulic system.

The platform has evolved throughout the project. Though it began simply, it has increased in complexity over three or four generations to accommodate the additional capabilities supported by the system, and required for more sophisticated prototyping. The most recent platform allows the system maintainers to swap different control systems with fairly short lead time.

Caterpillar has also done experiments with motion platforms, including a Frasca flight simulator, and a full-motion, limited visual, decommissioned B52 flight simulator at the University of Iowa Driving Simulator (part of the Center for Computer-Aided Design). They have done studies to determine the need for motion cues before integrating them with the *VPS*. In preliminary tests, they found that the ability to display “jerk” to the machine operator greatly enhances their ability to judge how the machine interacts with the simulated environment.

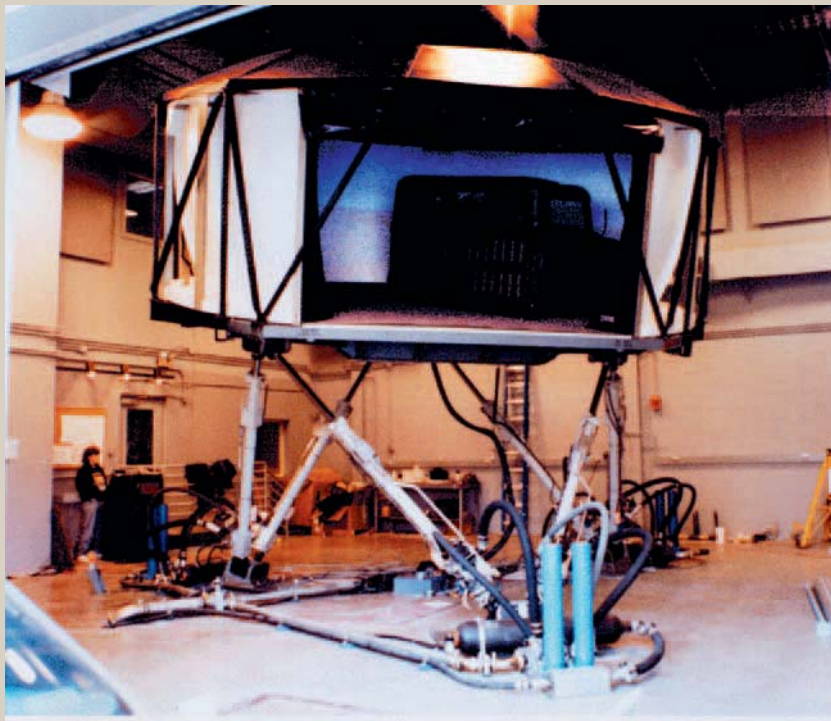


FIGURE 3-18

Caterpillar has experimented with a full motion flight simulator at the University of Iowa.

In the physical machines, the engine and hydraulic system produce many smells. These smells can actually be useful as a means of diagnosing trouble brewing with the machinery. However, no research has been done by the Caterpillar application designers along the lines of integrating such sensory display into the system.

Interaction with the Virtual World

The genre of this application is virtual prototyping. Specifically, virtual prototyping using a VR display system—not all prototyping done on computers in a virtual world requires the use of a VR display (an electronic circuit simulator, for example). The form is a participant vehicle pilot-through with control inputs from a platform mimicking the actual controls.

The narrative of this application is using the virtual machine in a typical work cycle. While the machine is put through its paces, machine-operator performance can be analyzed. Some sample questions the developers are interested in assessing are: How many passes does it take for operator to load a dump truck? How effective is the machine for the task at hand?

The travel interface for the *Caterpillar VPS* is simple. Control is a basic pilot-through interface that allows the operator to drive the machine as they would drive the real-world counterpart. However, it is not always appropriate for the user to interact in this way. There are many situations in which a user will need to be able to move around independent of the machine, perhaps while redesigning a particular part of the machine, or during collaborative sessions, while they watch other participants drive the equipment. For these reasons, it is also possible to move to a view outside of and untethered from the cab of the machine.

When traveling through the virtual world untethered from the machine cab, the *CAVE* wand prop is used. A pressure joystick mounted on the *CAVE* wand provides the user with two degrees of input. Pushing left or right rotates the world about them, and pushing forward or backward (toward themselves) translates them through the world in the direction they point the wand.

The system also has a menu system to control various options within the *VPS*. For example, operators can activate or deactivate sound rendering, or they may choose to jump

to another environment, such as a mining pit or highway road construction site.

The menu takes the form of an object that floats within the virtual world. The user can summon the menu by pressing a button on the wand. While viewing the menu, a marker indicates the currently selected item. This marker can be moved to the next item by making a quick downward gesture with the wand. To activate a submenu, a quick rightward gesture is performed. Once the marker is on the desired selection, another button on the wand is pressed, after which, the menu system is deactivated and the representation disappears. If no selection is made, then the menu representation remains in place in the virtual world.

Any of the menu options can also be invoked by using a keyboard equivalent shortcut. In general, the keyboard is outside of the *CAVE* environment, so keyboard commands are usually invoked by an assistant who is helping to run the *VPS*.

There are no computer-controlled agents in the *Virtual Prototyping System*. However, there are situations where such an agent would be useful. In particular, situations where a work cycle involves operations that must be performed in conjunction with operators controlling other equipment. For example, a dump truck agent may drive the truck away when it is full, with another truck driving into place.

A more current focus in the development of the *Caterpillar VPS* is support of distance collaborative capabilities. Caterpillar has design facilities located throughout the world, and requiring designers to congregate at a single location can be expensive in time and money. At the same time, not bringing designers together can allow less than ideal designs to propagate through the design process. By providing a shared virtual world in which geographically distant collaborators can see, speak, and work with one another while remaining at their home facility, *Caterpillar VPS* can provide a means of reducing the costs of travel, while enabling joint projects to gain the advantage of collaborative development.

Collaboration is seen as such an important aspect of working on a design project that the effort to incorporate multi-site collaboration capability within the virtual environment is considered a project unto itself—the *Distributed Virtual Reality (DVR)* project.

**FIGURE 3-19**

A menuing system allows the operator to change various aspects of the virtual world. The menu is summoned by pressing a button on the CAVE wand.

As human communication is both a verbal and visual operation, each site sharing the “conversation” is equipped with a video camera and microphone. The audio of each site is mixed with the overall audio of the virtual environment, and the video images are integrated into the avatar that represents the position of each site within the space. For sites without a video input, the avatar consists of a box-like shape colored based on the site it represents plus a matching-colored pointer that shows the position of their wand. When video is available, the avatar’s front side has the live images texture mapped onto it.

A multicasting approach was chosen for information sharing between sites involved in a collaborative gathering. Multicasting was chosen over a client/server model because of the reduced overhead of not needing to maintain a server application that duplicates most of the information already stored in the individual clients, and because

information between sites would all need to pass through a single server, which would result in increased latency. Other benefits of multicasting are that any site can begin interacting in the virtual environment first, and any other sites can join and leave the collaborative session at will.

The use of multicasting does raise some problematic issues that must be addressed, however. A major problem is that multicasting is not a fully supported feature over many networks. To enable a multicasting route between multiple sites requires special requests be made to the network administrators of all the intervening network “hops” to activate multicasting in their network routers. The other concern with multicasting is that it uses a data transfer protocol designed to keep network latency down by discarding old packets. Thus, when an application is too busy to read all the data, it will miss some information. In many cases, this is exactly what is desired. For example, only the



FIGURE 3-20

In the Caterpillar DVR project, participants at various geographically separated sites are able to share the virtual world in real time. Caterpillar engineer Bob Fenwick takes his turn in a CAVE that is a shared resource of the NCSA VR research facility. Avatars consisting of a box displaying a video of each person appear in the location and orientation the person is in the virtual world.

most recent information about the position of other participants is important. However, it is more difficult to handle situations where information regarding other collaborators joining or exiting the session is lost.

The data for the user and pointer locations, video images, and audio are transmitted from each site to the others over a multicast-enabled route over the Internet. The application also passes simulation information between sites. This allows the remote participant to see what actions the other participant has taken. The audio and video signals are captured using microphones and cameras connected to a computer, transmitted using standard network collaboration tools, and displayed by mixing the received audio with the simulator-generated audio, and integrating the video signal into the scene by converting it into a texture map that is rendered onto the avatar.

The goal of the world simulation is to reflect realistic performance of the machine and objects with which it interacts, in a real-time environment. The *VPS* uses a simplified transfer function from control inputs to speed, a simplified vehicle dynamics model, a simplified soil model, and a simple method for determining elevation for terrain following. Although Caterpillar has proprietary software that can simulate these systems very accurately in non-real-time simulations, they had to simplify the simulations to allow the required real-time performance in the *VPS*. Caterpillar used a proprietary dynamics simulator to simulate the various vehicle forces, including steering, hydraulic implement control, braking, and so on.

Venue

Design and testing of the *Caterpillar VPS* was performed in the research labs of the National Center for Supercomputing

Applications (NCSA) at the University of Illinois at Urbana-Champaign. Caterpillar later built a *CAVE* at their design facilities. Design engineers traveled to the VR facilities at NCSA in Urbana to spend 1 to 2 hours per session in the *VPS*. The engineers worked at their own pace subject to the availability of the *VPS*. The NCSA facilities are shared among several other researchers working with virtual reality, so the design team and users were required to schedule time to run the *VPS* in the NCSA *CAVE*. This scheduling constraint was lifted with the development of the on-site *CAVE* at Caterpillar.

VR System

The early implementation of the *Caterpillar VPS* used a Virtual Research Flight Helmet 2 HMD (aka “VR2”) and Polhemus Fastrak tracking system available in the NCSA VR facilities. A custom-built platform that emulated a backhoe machine was integrated into the NCSA environment. A large single-screen display was also available in the VR lab, and the developers also made use of it in the early application development and testing.

The most severe limitation of the original system was the low resolution of the HMD. It was evident early on that the resolution of the VR2 was insufficient for Caterpillar’s needs. During the course of early development, a *CAVE* system was installed at NCSA in collaboration with the University of Illinois at Chicago Electronic Visualization Lab. The NCSA *CAVE* was the second *CAVE* in existence, and Caterpillar quickly put the new display to the test.

The machine platform could be easily wheeled into the *CAVE*. The operator sat on the platform in the *CAVE*, and wore tracked Crystal Eyes shutter glasses. Since the original configuration of the NCSA *CAVE* included only a front wall, left wall, and floor, there was not an “appropriate” view for the operator’s right window. The NCSA *CAVE* could be configured to use the front and both side walls instead of the floor, but most of the other users of the NCSA cave found the two walls plus floor configuration more useful, so this was seldom done. In late 1995, with the upgrade to a new Silicon Graphics rendering engine (Onyx with dual Infinite Reality engines), the NCSA *CAVE* could more easily render to four separate walls, allowing for the addition of the right screen.

The latter system ran on a 12-processor SGI Onyx with two Infinite Reality Engines. The original NCSA VR lab was equipped with Silicon Graphics’ first multipipe rendering system, the dual-display SkyWriter™. The tracking system for the *CAVE* is a Flock of Birds electromagnetic system from Ascension Inc. The platform was originally connected to the VR system via an IBM PC equipped with an analog-to-digital input board connected to the Silicon Graphics via a serial connection. Later, an analog-to-digital board was directly installed into one of the lab’s SGI Onyx’s using a standard VME bus device.

In the original *VPS*, a Symbolic Sound Kyma system was used to play triggered audio samples. The current version uses the NCSA VSS audio system to play triggered audio samples directly from a Silicon Graphics workstation. Prior to sound hardware becoming available on the powerful Onyx series of machines, an SGI Indy workstation served as the audio serving device.

Application Implementation

The *Caterpillar VPS* and *DVR* projects are ongoing research-and-development efforts. The projects began as a collaboration between Caterpillar and the NCSA Visualization group in 1990. The *VPS* was developed by Caterpillar, NCSA, and the Electronic Visualization Lab at the University of Illinois at Chicago. Work on *DVR* formally began in September of 1995, with functional experiments running in July and August of 1997.

The project represents many “mythical person-years” of development. Project personnel have included project leads Rich Ingram, then Kem Ahlers, and Bob Fenwick, and project developers Dee Chapman (NCSA), Mike Novak (NCSA), Valerie Lehner (NCSA), Brian Keenan (NCSA), Lance Arsenault (NCSA), Volodymyr Kindratenko (NCSA), Erik Zentmyer (NCSA), Joe Lohmar (NCSA), Mike McNeill (NCSA), and Milana Huang (EVL).

In the first version of the system, very simple scenes were rendered using a research graphics toolkit from the University of Iowa. The only parts of the virtual world rendered were the cab frame and digging arm of the backhoe loader. The interior of the cab was not visible, including the control levers.

**FIGURE 3-21**

In order to provide a more useful experience, Caterpillar modeled several testing locations. The image above depicts mining pit, and the image below depicts a strip mine.

The Front End Loader was developed using the University of Iowa toolkit, but later, the project developers decided to move to a more robust toolkit with built-in capabilities for virtual reality. They chose Sense8's *World Tool Kit (WTK)*. The graphics are computed with a combination of *WTK* and portions of the *CAVE* Library. Once the switch was made to the *CAVE* environment, the HMD specific work was abandoned—although the *WTK* and *CAVE* libraries both allow for the possibility of rendering to HMDs.

Conclusion

In addition to the direct benefits to Caterpillar from the use of the *VPS*, they also enjoyed a great deal of publicity about the project. The public is very interested in the use of high technology in development of products they know and understand. The Caterpillar project has aided in allowing the public to get an early glimpse of how virtual reality can be put to very practical uses. In 1993 Caterpillar Inc. was awarded NCSA's Grand Challenge Award for their work on the *Virtual Prototyping System*.

The fact that actual design engineers used the system is an encouraging indicator of the utility of the *VPS*. The Caterpillar® 914G Wheel Loader is one example of a released product which used the *VPS* in its design. One of the benefits attributed to the *VPS* is enhanced rear-view operator visibility. This influenced a major design change in the product line. Engineers used the *VPS* in the development of other products and to make major design changes to existing products that had not been released yet.

It is not clear to what degree user's become "immersed" in the virtual environment presented by the *VPS*. However, designers frequently made trips to the NCSA *CAVE* specifically to try new design or control interface ideas because of the realism that they simply could not achieve with a desktop computer interface. On the other hand, in instances where the test operator needs to have a good feel for how the unit drives over rough terrain, a motion-base display is almost essential to provide an appropriately immersive experience.

Over the course of the years during which the *Caterpillar VPS* has evolved, the developers have learned many lessons about implementing a large, collaborative virtual reality system for a design environment. The first major lesson was the necessity (and difficulty) of converting the CAD files used in product design to a format more suitable for real-time VR rendering.

Many of the other lessons learned involve the integration of collaborative capabilities into a virtual environment. The developers found that low-latency audio was essential, but more easily producible than video. Telephone-quality (8 KHz) audio and 8 Hz video were sufficient and producible within the virtual world. The use of standard multicasting protocols and tools proved practical and suitable as the means of adding a collaboration interface. However, challenges still existed regarding the difficulties in acquiring a high-bandwidth, low-latency, multicasting route between sites (especially trans-continental sites).

Caterpillar is also interested in obtaining customer opinion on potential future product designs and design alternatives in products that have not yet been released. They may ask customers for opinions on joystick versus steering wheel methods of steering control, for example. Connecting these interfaces to a virtual representation allows them to place the new concept on any of Caterpillar's virtual machines operating in a variety of settings. Input controls of modern machinery are a "drive-by-wire" control system. This means that it is now easier to replace or change how the input controls affect the machinery. Putting new devices through human factors studies before they are integrated in physical machinery is an appropriate use of VR.

Finally, Caterpillar continues to see potential in using VR in training operators to drive machines with which they are not familiar in a safe environment. Often, the operators must use the machines in rugged or hazardous conditions. Caterpillar is making every effort to teach operators how to operate machines safely in such environments.

Bob Fenwick of Caterpillar summarizes the *VPS* effort:

Even though Caterpillar has used virtual reality in product design for a number of years, we still consider this to be an infant technology. Its ultimate potential will not be realized until it becomes an integrated everyday tool that product design engineers use in the new product development cycle. Advances in hardware, software and communications brings us a step closer to that realization every day.

CASE STUDY 3.3: GENERAL MOTORS CORPORATION

Executive Summary

General Motors developed an in-house Virtual Reality Facility as an extension of their *VisualEyes*[™] project. The *VisualEyes* project enables them to do virtual prototyping of new vehicle designs prior to creating physical prototypes. An important aspect of the design process was to be able to see models at life-size. To enable this, General Motors utilized *CAVE*[™] and other stationary projection devices. The *CAVE* system allowed designers to view vehicle interiors from the interior perspective. They also used large, single-wall displays with no user tracking to display crash test data, which was best seen from outside the car.

A primary goal of the GM VR project was to display the models with a very high degree of realism. Thus, they chose to render highly detailed images at the expense of high frame rate and interactivity. Audio was not a critical element of this application, and the developers planned to enable collaboration between design teams located at geographically disparate facilities.

Introduction

General Motors (GM) has a history of early adoption of computing technology. They are credited with the development of the light pen, and are among the first users of computer-based solid modeling. Thus it is no surprise that they are among the first businesses to investigate the use of virtual reality toward improving products.

GM virtual reality efforts fall within the scope of their *VisualEyes*[™] project. The primary purpose was to allow design engineers to see a visually realistic, life-size model of their design in an immersive environment. The *VisualEyes* package refers to a software program developed by GM to aid in virtual prototyping ranging from the desktop to large-screen and immersive displays.

The old adage “faster, better, cheaper” describes some of the primary goals of this application of VR. Any technology that offers the possibility of more rapidly bringing a better product to market attracted the GM researchers. Virtual reality, via a *CAVE* display, was their chosen venue.

One of the more explicit goals of the *VisualEyes* project was to reduce (or eliminate) the time and money spent on

producing physical prototypes of a future product. Physical prototypes are very costly, and in iterative design processes, many are built. If the number of physical prototypes constructed can be reduced, especially in the early design stages, a considerable cost savings can be realized. It is also the case that physical prototypes take time to construct, so reducing the number that need to be constructed results in a faster path to a final design. These concerns are very similar to those experienced by Caterpillar Inc. and Boeing Corp.

Interestingly, the goal at GM focused on visual realism in a virtual prototype, even if it came at the expense of response time and interactivity. In order to use the virtual prototype in design evaluation, they required a high-quality rendering to allow evaluation of overall impact of design, visual quality inspection, human factors, and packaging studies. While they were also interested in using interactive tools for the added ability to view a model under different lighting conditions and from different perspectives, they directed most computational resources toward creating a realistic rendering.

Without the virtual reality system, design engineers make use of a variety of real-world physical models. In the earlier stages of the design process, they used clay models, usually scaled down in size, but occasionally life-size clay models were created. Although much more costly and time consuming, life-size models were created because people are much more adept at understanding and reacting to “models” that are full scale.

Designers have also become accustomed to viewing computer graphics renderings of their designs on their desktop workstations. The *VisualEyes* software renders to a variety of visual displays ranging from desktop viewing to a *CAVE* projection-based VR system. Hence, the virtual reality aspect of the project was part of an exploration into an alternative method of viewing virtual worlds previously viewed on conventional desktop monitors.

Finally, in traditional design practice, full-size prototype physical models are still constructed. The cost of these models can be prohibitively expensive, so they are utilized only when absolutely necessary. However, they do provide a very accurate representation of the visual aspects of the

**FIGURE 3-22**

General Motors chose the CAVE to examine the aesthetics of an automobile interior.

final product. The accuracy of the representation allows designers to uncover most problems in the design.

The paradox is that by the time the stage of design is reached, it is difficult and costly to make drastic changes. *Virtual prototyping* allows an intuitive way of exploring new designs. In GM's case, they were more interested in finding problems rather than validating the design. Virtual prototyping allows this to happen much earlier in the design life cycle, yet still provides a representation that visually emulates the final product.

Application Description

GM's initial focus on using VR was to render a surface model of the interior of a vehicle for evaluation of the aesthetics and ergonomics of a design. *VisualEyes* already provided an acceptable means for lifelike viewing the exterior of a virtual car, a large-screen wall projection. Such a display matches how one naturally views a car: from the

outside; standing next to it; looking at one side at a time. Viewing the interior of a car, however, is not typically done in this manner. To get a good feel for the interior of a car, one needs to be inside. This means that the user is able to turn their head to see what the interior looks like from any perspective.

Virtual reality capabilities were integrated into the *VisualEyes* package to enable the proper viewing and design analysis of automobile interiors. By positioning the viewer in different locations in the car, they can evaluate the interior from the perspective of the driver, or any of the passenger seats. Having a physically immersive display also allows users to do things such as poke their head under the hood to see the interior of the engine compartment.

One of the key points of the system is that it provides a life-size view of the car. This allows the engineer or designer to exploit their past experiences of sitting in many actual

cars. One designer exclaimed on seeing a computer screen display, then a life-size immersive display: "That is a model, but *this* is a car!"

Representation of the Virtual World

The virtual world for design analysis is a simple one, a highly detailed visual model of a car. This is sometimes enhanced with a showroom or outdoor scene in which the car is placed, but the car is the important element. As a visual design tool, aural elements in the scene are not very important, with the exception of feedback from the control interface. The virtual world display is often augmented by the use of physical props such as a seat and steering wheel.

In many VR applications, the choice of visual representation is not obvious. Considerable thought goes into how things should be represented, such as which symbols or graphical elements can be mapped to particular components of the virtual world. In this application, however, the choice of representations is simple. The virtual world

should be represented as realistically as possible, a lifelike representation of the automobile.

In many virtual reality and computer graphics applications, datasets are simplified in order to increase rendering speed. In this application, with the explicit goal of attaining as much realism as possible, data simplification is avoided unless absolutely essential. Models are represented within the computer as polygonal objects, as well as texture maps. Standard culling techniques based on distance are not useful in environments such as a car interior, where everything is nearby. This problem also occurs for other types of information-visualization needs, such as many scientific visualization representations.

Frequently, the objects in the virtual world can be distinguished between the object of interest and background objects. One technique they use to help maintain more reasonable frame rates is to direct more computational resources toward the focal object, presenting it in full



FIGURE 3-23

A designer views the interior of a car through the window in a CAVE display. Note that the virtual car is life-size.

detail, while spending less on the background objects. The background objects may be rendered with less polygonal detail, as points, or in wireframe. Another option is to

simply not display the nonessential objects. Their highest concern is to maintain the aesthetic integrity of the object of importance.

DESIGN CHOICE

In this application, realistic rendering was done at the expense of frame rate, rather than model simplification. Compared with standard VR practices, this decision was carried to the extreme, delivering frame rates that would generally be considered quite bad or unusable for out-the-window visual simulation—often approximately 5 frames per second (FPS), but sometimes as low as 1 FPS. For visualization of a static scene, low frame rates are preferable to cartoonish simplifications of higher rates.

To achieve a very realistic-looking display, details are very important. For the interior of a car, the most critical element is the dashboard display.

In some instances, texture maps can be exploited to aid in presenting a lifelike appearance. A variety of different texture map techniques are used in this application. Some of the textures are photographs of actual components of a car. Others are hand drawn. In special cases stereographic texture maps are used. In general, for VR, stereo texture maps are not useful because they must be viewed from a particular point of view. In this application, however, the developers had the luxury of knowing the most important and most common point of view (the driver's seat) and can create the stereo textures appropriately.

Although the system did model the mirrors in the interior of the car, the participant in the simulation does not see

their own reflection. A mirror view could be accomplished by using a video camera and texture mapping the video image onto the mirror surface, but this does not contribute significantly toward the goal of design analysis.

This particular application focused on the visual aspects of design. There was no audio representation, though feedback at the control interface could have been integrated.

Haptically, the application involved a participant sitting in an actual car bucket seat, giving the strong sense they were sitting in a car. Ideally, the best choice of seat to use was found to be one from the car being designed. However, depending on what stage of the design process the car was in, a physical copy of the seat itself may not yet exist.

Allowing the user to hold and manipulate an actual steering wheel, or a steering wheel prop, also raises the sense of realism. Since vehicle dynamics were not part of this evaluation, the steering wheel did not provide any input to the system. However, its usage was enough to improve the perception of the design.

VisualEyes was used to simulate a stationary car; there was no need for vestibular display as might be needed for an off-road driving simulator. Likewise, the senses of taste and smell are not important to the task (though using “new car smell” aerosols may add to the mood.)

Interaction with the Virtual World

This application is of the genre *virtual prototyping*. In virtual prototyping, an ephemeral model is put before the designer for evaluation. The main method of interaction (the form) is simple—physical locomotion. There are also menu interactions given via a wand that can be used to select the make and model, or modify the color of the body or interior. The focus is to let the designer naturally

move about the space within the vehicle to make their evaluation.

The “narrative” is typical of design and visualization applications—look at some pre-created data from different perspectives, and with some user-selectable options.

The best way to determine the quality of a vehicle design is to interact with the representation in a manner as similar

as possible as one would with the final product. This means climbing in, sitting down, reaching for the radio controls, etc. Thus physical movement (locomotion) is the travel method best suited for this application. The application developers also implemented other travel options, including fly-through methods. One option that is sometimes used to interact with the design prototype is the *grab-the-world* method of travel.

Most of the interaction that takes place in immersive use of the *VisualEyes* system took the form of physical body movement to judge the appearance and human factors issues of a design. For some design analysis situations some additional interactions were available, such as being able to choose the style of the interior dash, or to choose the interior and/or exterior color scheme. For these interactions, a simple menu-based interface was provided using the buttons on a handheld wand.

The *VisualEyes* software system as described here does not have built-in support for remote collaboration. The system was developed in the GM research and development center

in Warren, Michigan. During the initial development states, designers went to the R&D center to try *VisualEyes* on their designs. Later it was exported to the Warren Design Center.

The *CAVE* itself is ideal for multiple people at the same physical locale to work together on a design. One reason for choosing to use a *CAVE* and large-screen displays is the ability for many people to view the environment simultaneously. This means that there is opportunity for people to share ideas while experiencing these images, all in an immersive environment

The virtual world in this application was a static model. Car designs are examined as they remain motionless in the virtual world with the occasional exception of simulating the car placed on a rotating base in a virtual car showroom.

There is no collision detection done within *VisualEyes*. This allows the participant to do things such as stand up in the car, and enter and exit through the side panel. There is also no notion of gravity in the system, but there typically aren't



FIGURE 3-24

A life-size rendering allows a more intuitive visual interpretation of simulated crash data.

individual parts to pick up and move, so this is not a major consequence.

Another use for *VisualEyes* is visualizing computational simulations generated for crash analysis. Crash analysis rendering is typically displayed on the nonimmersive (untracked), large-screen display as this data is more suitable for observing the car from the exterior. The crash physics themselves are pre-computed, and not part of the rendering package.

Venue

The venue for this application was a research lab at GM. Another possibility was to use the system in dealership showrooms, allowing customers to experience a virtual replica of a car they are considering with various options, color choices, etc.

At the GM R&D center, the normal scenario was for small working groups of 5–6 people to visit for sessions of about 30 minutes to 1 hour. These working groups were often subsets of a larger group who meet around a table, then proceed into the *CAVE* to discuss some aspect of the design. The car designers typically used the shuttering glasses less than 30 minutes at a time.

In addition to actual design meetings, many demonstrations of the system were held. This is a price that system designer Randy Smith says is required for promoting a new technology.

VR System

VisualEyes was first used to display models in stereo on large monitors using head tracking and hand (wand) tracking with Logitech ultrasonic tracking systems. Later, *VisualEyes* was used on larger projection screens to view larger models of full-sized cars that were sometimes head-tracked, sometimes not.

When displaying cars and parts on a small screen with head tracking, the developers found that designers were not particularly interested in looking at the cars, but reacted significantly when shown a (smallish) part at life-size. Thus, they strove to push their displays to be large enough to display a car at actual scale. Providing the user a life-size view of the exterior of a car worked well, even when head tracking was disabled. For this purpose and for viewing visualizations of crash simulations, untracked rendering was satisfactory.

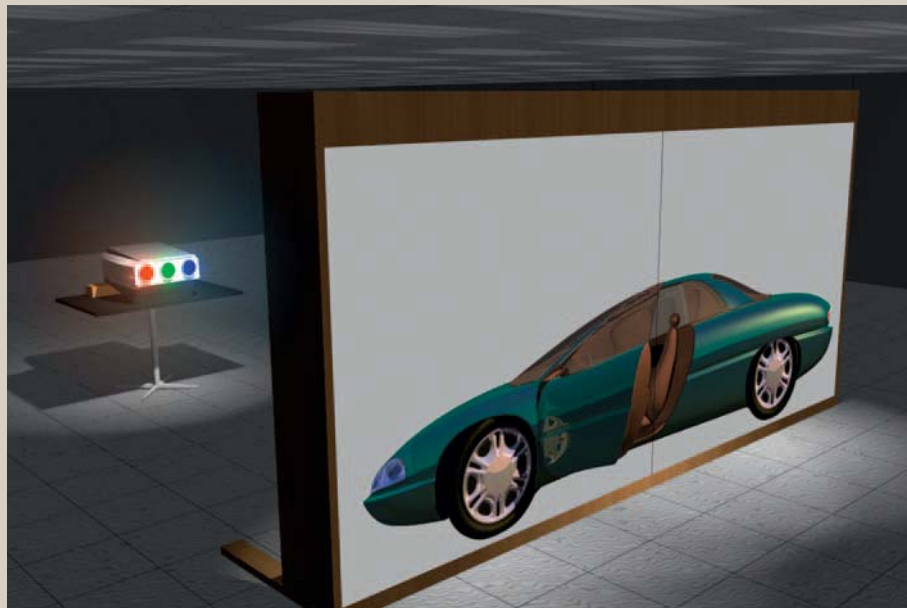


FIGURE 3-25

Rear projection of a vehicle allows designers to inspect the image without casting shadows on the screen.

To examine the interior of a car requires the ability to see all around. Thus, *VisualEyes* was extended to work in a *CAVE* environment, a visual display that surrounds the user. Installed in 1994, GM's was the first

CAVE at an industry facility. From its creation, their *CAVE* offered a full three-wall display plus the floor display. Lack of a rear projection wall was not a major hindrance.

DESIGN CHOICE

From the outset of this project the application designers had planned to use projection VR rather than an HMD-based system. There are a variety of reasons for this choice. One of the major reasons is that typical usage would involve a team of people discussing a design. Projection VR allows team members to easily see and converse with one another. Another reason is that the developers felt that HMDs were too heavy and clumsy,

and many of their clientele, including executives who are interested in viewing designs under development, would not wear them. They also felt that the HMDs available (when the decision was made) were not adequate with respect to resolution and field-of-view. GM was quite satisfied with the choice of projection VR, and even if given adequate resolution and field-of-view, it is unlikely they would shift to an HMD system.

Interestingly, they chose to build a *CAVE* measuring 8 feet on each side as opposed to the more common 10 feet per side cube. This was done to more closely map to the size of a car interior, minimizing the disparity between focal distance of the model and the physiological measurements calculated by the human optical system. Thus the screens (where the eyes will focus) are approximately the same distance from the participant as the interior wall of the car. Secondary benefits are that by reducing the size of the screens, the apparent resolution (pixel density) of the display is increased, less floor space is required, and the display is brighter.

In comparing the *CAVE* environment to the large, single-screen display used for the crash simulations, the application developers said that they believe the advantage of the *CAVE* was that it allows the inside-out view that gives the participants a chance to climb inside a crashing car, or to poke their head into the engine compartment. The advantage of the large single-wall projection was that it is more suitable for external viewing of a vehicle, and allowed even larger groups to watch the simulation.

The application designers' primary concerns with the projection-based system were insufficient resolution, brightness,

and color uniformity across screens. They wanted improvements in each of these areas. An upgrade to SGI Infinite Reality Engines provided an increase in pixels per screen from 1024×768 to 1280×1024 (66% increase), while continuing to provide a 96Hz refresh rate (48 Hz per eye when in stereoscopic display mode). This has resulted in a noticeable improvement in visual quality, but they felt like they were an order of magnitude away from where they would like to be. For example, lines on dashboard gauges were in reality 1 mm wide. On the old system, they were only able to resolve 2–3 mm, which was not quite good enough for presenting a realistic display.

Like most *CAVE* displays, the GM *CAVE* used the Ascension Flock of Birds extended-range electromagnetic tracking system to track head position and the position of one handheld device. Because electromagnetic tracking systems are susceptible to errors caused by metal in the environment, tracking performance is improved by calibrating the rendering system to correct for the inherent errors. At GM, they used a mechanical position measuring system from FARO (the FAROarm™ to measure the difference between the values reported by electromagnetic system and the (more accurate) values from the

FAROarm. These comparisons were the basis from which the tracking system is calibrated. The FAROarm itself can serve as a tracking device, and such an option is available in *VisualEyes*. This is very useful when recording the virtual environment with a video camera. The camera was attached to the FAROarm, so the perspective is rendered from the camera's point of view.

The computing and graphics system utilized an SGI rack Onyx with Infinite Reality graphics engines. The Onyx system included multiple processors enabling it to render and compute interactions in parallel. This was the typical system for a *CAVE* environment at the time.

Application Implementation

This project was developed around the already existing *VisualEyes* software environment. Rather than rewriting their existing software to work with the *CAVE* library, the application developers integrated the basic functionality of the *CAVE* libraries directly into their existing software, providing the ability to display head-tracked, stereoscopic images in a multiwall display. By integrating the multiwall display techniques directly into their existing system, the application developers felt they were able to reduce time of software programming and maintain the use of the advanced rendering techniques they developed for their visualization system.

GM began exploring VR in 1992 as part of a continually evolving company-wide internal project. By 1992, they felt that the supporting technology for VR had finally reached the threshold to make VR a usable medium for their purposes. The GM VR project was a boot-strapped effort that grew as necessary. They evolved the system to fulfill the need for an inside out view of full-size models, stereoscopic depth cues, and tracking to make the perspective accurate.

The *VisualEyes* project was developed in-house, within the GM Research and Development Center by a team consisting of Randall C. Smith, project creator and team leader; Tom Celusnak and Larry Peruski, also early members of the core software team; and Don McMillan, who was responsible for some of the more recent developments.

By corporate standards, the development and usage of the system proceeded rapidly, though the designers wanted to move even faster. Gaining buy-in from the users and company required time, demonstrations, and energy.

The typical mode of interaction with users and potential users was to tell them to BYOD, or "Bring your own data." Usually the design data came available in a CAD format that could be readily transferred over the company network, and imported into the system. Using the participants' own data allowed them to better understand the capabilities of the new system. In fact, they saw many repeat visitors who came to see their designs in the *CAVE* environment.

Conclusion

An interesting anecdote is that people reported that virtual cars viewed in the *CAVE* seemed wider than expected. A possible explanation of this phenomenon is that it is a result from the fact that the *CAVE* was slightly wider than a typical car, and eyes focus at the distance of the *CAVE* wall. The brain uses this information to help determine the size of the interior, resulting in the wider perception. The focal distance depth cue may be overridden by stereopsis⁶ when directly looking at the walls of the car, but when the head is facing the front of the car stereopsis is not a factor because each side-wall can be seen by only one eye.

When examining the car from the driver's seat, the viewer sat about 2 feet from the left wall (driver-side in the United States). This put the passenger-side wall about 6 feet away from the viewer, which caused the single right eye to sense the distance of objects displayed on that screen as being 6 feet away, which is larger than most cars.

Although *VisualEyes* probably has not directly saved any lives,⁷ it is likely that it saved GM time and money. GM has not

⁶Lack of stereopsis in peripheral visual is not the result of projector or head-based VR displays, but rather is an inherent trait of the human visual system.

⁷It is, however, used in crash simulation analysis to make safer cars.

published any data to verify whether any time or money has actually been saved, but the longevity of the system implies that was indicative of payback in the near future. Although the system was used on real problems within GM (indicating that the system is useful to designers), users do report that there are additional capabilities they would like to have.

The developers state that the system probably allowed the designers to reduce the number of early physical models resulting in some monetary savings. However, they did not reach the point where construction of all physical models could be eliminated. This is particularly true for the end of the design cycle, where physical parts are at least required for use in manufacturing analysis and crash testing. Government agencies are not yet to the point of trusting computer simulations enough to alleviate the need for physical crash testing. But any reduction in the number of stages requiring a physical model results in significant savings of both time and money, which can add up to considerable savings.

Although participants in this experience probably don't feel like they are actually in a real car, the application designers did relate some anecdotes that indicate participants' sense of immersion, and the transparency of the user interface.

One anecdote involves an executive who was examining a design for a specially designed ride vehicle for display at a future technology venue at Disney's EPCOT center. The executive looked carefully at the vehicle, squatted down, sat in the seat, etc. He remarked that the carlike ride vehicle required too large a step to get in, and worried about how people's heads would block others' views. The interesting aspect about this story is that the executive was not remarking on, or testing, the technology. He was doing his job, and the VR system supported that task.

Other anecdotes include the excitement participants display in the system, and the fact that many would hesitate to step on the console and tended to reach to set their notebook down on the dash, etc.

One of the more significant things that this group has learned is the importance of detail over frame rate in their application. In other words, sometimes using sufficient detail makes the system much more useful even if it costs a rapid display update.

Having begun on the path of projection VR, the application designers did not regret that decision. The application designers felt that projection gave them the ability to see and more easily work with collaborators. Also, there was no need for suiting up, and the resolution and field-of-view were better when using projection technology. They also learned that VR was not a suitable technology as a tool for fine modeling due to tracker error and low resolution.

In turn, GM learned that people who are not designers by trade typically have difficulty visualizing from a drawing or a sketch. It turns out in practice that what they had built primarily as a design tool actually turned out to be primarily a communication tool.

In the future, the application developers envisioned using the system for worldwide collaborative efforts to reduce travel time and expense. Design discussions can be held between collaborators working on different continents to work together more frequently and cohesively. Designers have specific concerns related to their location. Many of these different concerns might not be raised as early in the design process if meetings are held only at times when they could meet in person. In general, fostering early communication during design helps the team to find out about the little things that will be major issues down the road. It also helps keep continuity during the design process. VR is becoming part of GM's virtual prototyping effort. Application developer Randy Smith likes to quote an old Chinese proverb to describe the usefulness of VR: "If I hear—I will forget if I see—I may remember if I do—I will understand."

CASE STUDY 3.4: NALCO FUEL TECH

Executive Summary

The BoilerMaker application was a joint development project between Nalco Fuel Tech (NFT) and Argonne National Laboratory (ANL). This collaboration created a virtual reality application in which one can interactively do visualization, analysis, and design of significantly reduced-emission boilers and incinerators. The *raison d'être* of this application was to provide a way to find the optimal locations for fuel injectors in an industrial boiler. By accomplishing this, NFT augmented the pollution control systems of industries to help them meet the requirements of the 1995 Clean Air Act. The *BoilerMaker* application was designed to aid and assist designers for placement of fuel injectors in their boilers. This application allowed designers to view simulation data and interact with the changes that varying injector placements resulted in, especially temperature gradients and effects of design changes.

Application Description

Placing boiler fuel injectors in locations that result in optimal (cleaner) burn is a difficult, iterative task. The optimal location is influenced by flue-gas temperatures, velocity fields, and slag buildup. The application authors created a computational model of the interaction of these elements that allowed them to predict the effectiveness of an injector at a specific location. The VR application gave them a tool to interactively place injectors and get quick feedback of spray coverage and burn efficiency. By interactively moving the injector nozzles and particulate matter in the virtual boiler, the designer could quickly iterate over several placements to arrive at ideal locations.

The heart of this application is the computational model that simulates the fuel-burning phenomena of a boiler. A separate software package (*TrackPack*) computes the

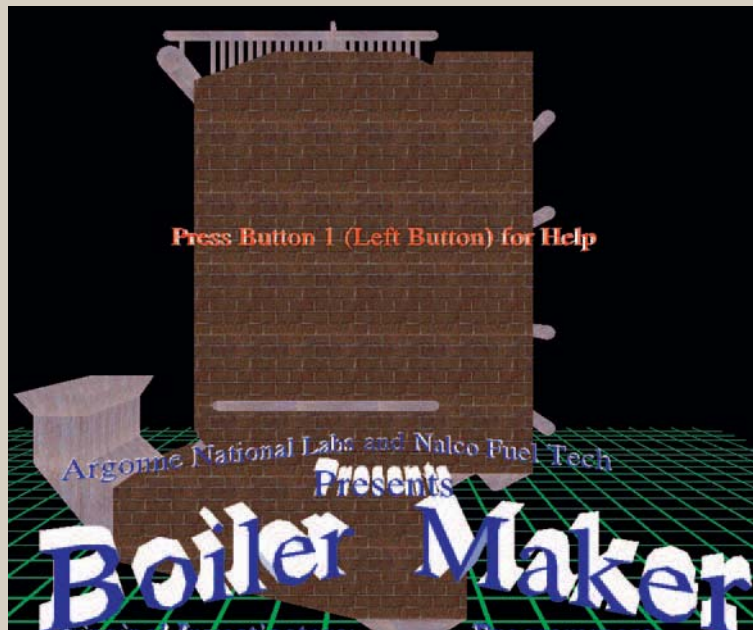


FIGURE 3-26

Participants begin their experience with the Nalco Fuel Tech BoilerMaker application by making menu selections via a CAVE wand while standing outside a virtual boiler.

dynamics of particulate matter. Simulated particles can be massless, massed, or evaporating, and calculated on a variety of meshes for both steady-state and time-dependent flows in noncoupled two-phase flows. Although the model is computationally intensive, it does respond adequately for real-time interaction—20 frames-per-second (FPS) for moving an injector, while the computation of a handful of injected particles proceeds at about 2 FPS. By distributing this application over multiple machines, each computer can do the part for which it is best suited (parallel computation/graphics rendering), resulting in a useful tool. The display and interaction with the simulation is carried out in a *CAVE* equipped with a wand and electromagnetic tracking system.

The utilization of networking hardware to allow running an application in distributed mode has both advantages and disadvantages. The primary advantage is that each component can run on a system especially suited for a particular computation. The main disadvantage is the added complexity of building a distributed system, as well as the need for high-speed networking to connect the components. When the *BoilerMaker* VR experience begins, the user is located above the ground (a grid of green lines) facing the external structure of a boiler. Interaction with the environment is accomplished via wand-based travel controls, and a menu interface developed by ANL. Basic operation includes three modes of interaction:

- Display features mode
- Visualization mode
- Injector placement mode

The display features mode allows the user to increase frame rate by toggling the display of graphically intensive geometric features. For instance, the user can toggle the display of the internal water and steam pipes representation. These cylindrical graphical objects are made of many-thousand polygons, and put a strain on the rendering system. This noticeably reduces the visual frame rate of the application. By removing this visual feature, the frame rate jumps to a very comfortable level—about 40 FPS even when flying vectors are in use—making interaction much more effective.

The visualization mode allows a user to display a variety of common visualization idioms including a streamline from a point, an isosurface, and static or flying vectors.

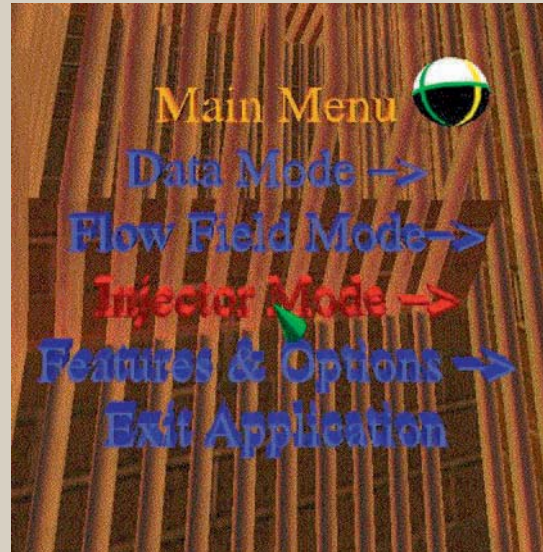


FIGURE 3-27

The BoilerMaker application presents a menu of three modes of operation: the small sphere in the upper right is a three-dimensional compass to aid the participant in determining their orientation in the virtual world.

Streamlines are released using the wand location to position the release point, and a button press to activate. An arrow loops over the path of each streamline showing the speed of flow along the path. Isosurfaces can be used to visualize a scalar data field such as the region for which temperatures are within the range for highly efficient particle burn. The overall flow of gases can be viewed with a collection of vectors (represented as darts). These darts can be visualized either as a static collection covering the entire boiler, or as a smaller collection that traverse the flow of the boiler beginning at the fire.

Any of the visualization idioms can be colored based on different scalar fields within the simulation, primarily temperature and optimal burn. Coloring for optimal burn is also a temperature-based mapping, but highlights the region where the temperature is ideal. Specifically, bright colors are mapped to the optimal region, light purple to regions that are too cold, and dark purple to regions that are too hot.

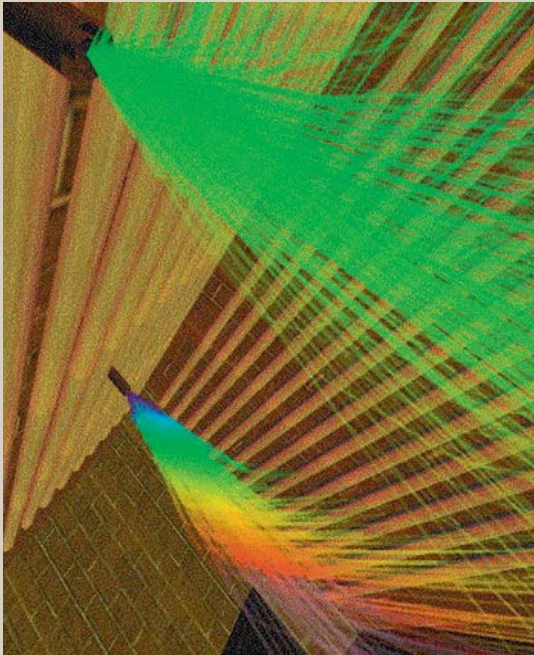


FIGURE 3-28

Colored stream lines indicate the effectiveness of the fuel injector placement. Here we see that light purple regions show areas that are too cold for effective burning, whereas dark purple indicates regions that are too hot. The very bright areas indicate optimal burn temperature.

The injector placement mode allows the user to interactively position up to 25 injectors by pointing at a location on the boiler wall, selecting an existing injector or creating a new one if none are nearby. By pointing at exterior walls, the wand is then used to drag the selected injector to the desired location. As an injector is moved, computations are performed to aid in visualizing the effective burn of the current position. Characteristic sprays are made from the injector, showing calculated burn of entering particles. The participant uses the visualization to place injectors at locations appearing to have effective burn efficiency.

The *BoilerMaker* user interface also provides for computational steering. Computational steering is the term used to describe controlling a running computational simulation

from within a virtual environment. In addition to the menu interface, widgets such as 3D scrollbars allow the user to modify the numerical model parameters and visualization parameters. The initial particle size, speed, and distribution from the injector source can be adjusted using the scrollbars when studying particulate matter.

Audio is used to enhance the feedback to the user. The user can hear the transition between being outside the boiler and being inside the boiler by a marked difference in the volume of the ambient fire burning rumble. A trumpet fanfare marks the event of a new dataset being received from the computation application. Also, each release of flying vectors into the flow field is marked by a swoosh sound. However, no audio feedback is provided for user interface features such as the menu or controls relating to the travel interface (which would be useful).

The method of travel in this application is fly-through, with no constraints on orientation. This is unlike many applications that allow three-dimensional flight through space but constrain the view to have a specific up vector (i.e., if standing, the ground is always in the direction of the feet). In *BoilerMaker*, the user can rotate to any orientation with respect to the world. Such freedom can lead to disorientation. To compensate for this, they introduced a “3D compass.” The compass is displayed at a fixed position relative to the user. Drawn as a sphere, it uses colors to disambiguate directions. The most dramatic distinction is between up and down. The top hemisphere is white, the bottom black, giving the appearance of an airplane pilot’s attitude indicator with the equator acting as the artificial horizon. The four vertical meridians are colored red, green, blue, and yellow to help the user determine their orientation.

The developers’ initial goal for travel was to design something that emulates a flight simulator. Because the user can hover, the resultant method is perhaps more akin to a simplified helicopter interface.

Travel control is divided into two components: rotational and longitudinal motion. Rotation is controlled by rotating the wand about the wrist. The Y-component of the joystick on the wand is used to move longitudinally through the world (along the line the user is facing). The participant can lock travel by pressing right on the joystick. Pressing left reactivates travel control. Locking travel means that

both rotational and longitudinal movements are deactivated, freezing the user at the current position.

This travel technique is a little complicated but can be mastered in a single session, and is quite effective, with only a couple of small difficulties. One problem is that there is no way to directly move laterally other than physical locomotion. The other annoyance is that because the travel-lock control is part of the same device as the speed control, there is a tendency to activate the freeze mode when turning right. This is exacerbated since there is no aural feedback to indicate when this happens.

Once in operation, a typical design process using *BoilerMaker* begins with the application experts constructing the 3D and computational model of the boiler (or incinerator). This step can be done in about one half day. They then used the *BoilerMaker* virtual reality application to explore the gas flow and efficient burn regions of that particular unit. Once they had an understanding of the burn process of the unit, the engineers begin to design the injection system. This step generally took several iterations, and without VR is the most time-consuming phase. Then, if necessary, the *BoilerMaker* application can be used again (with remote collaboration) if the on-site engineers encountered an injector that cannot be placed as planned.

An avatar consisting of simple body, head, and hand representations allows a user to determine where another participant is looking and pointing. While this can be effective for collaboratively exploring the data of a system under design, it can be even more effective for working on last-minute changes that are found to be necessary during the implementation of the design. In particular, if the engineers on-site at the installation of the actual boiler discovered that placement of an injector was not possible as planned, they could use a more portable system to collaborate with the (off-site) design experts to determine a feasible alternate location. The experts at the home office would have the advantage of more powerful computers and better VR display systems. Thus, they could more quickly evaluate other possible injector placements and show the result to the engineer on site at the boiler/incinerator.

VR System

This application was developed for use in a typical CAVE installation, using the *EVL CAVE* library. However, the *CAVE*

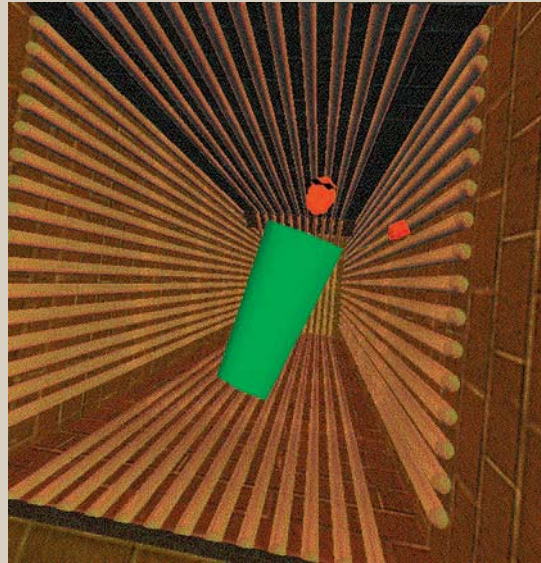


FIGURE 3-29

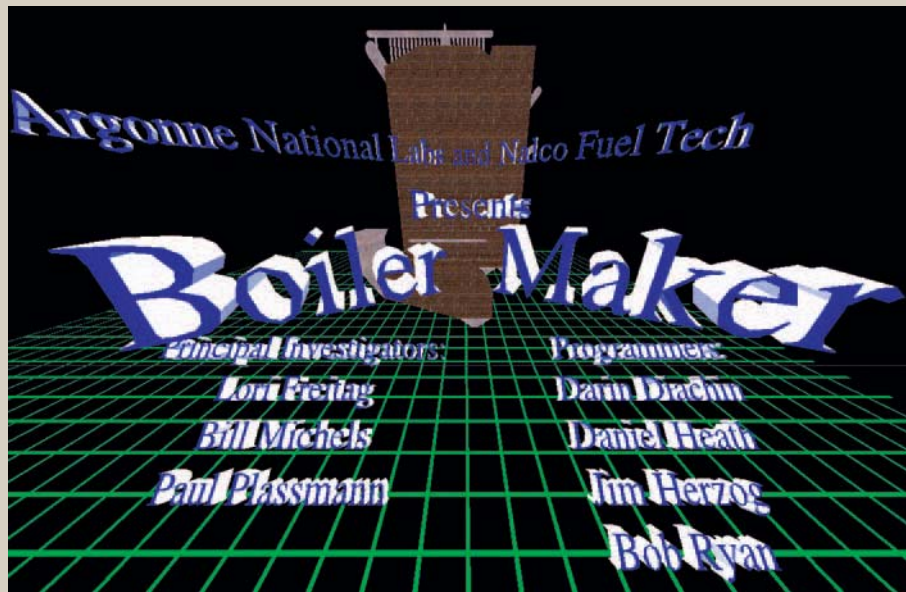
The BoilerMaker application supports synchronous collaboration between multiple sites. Here we see the avatar of one participant exploring the pipes in the Argonne's CAVEComm library, used to enable distributed computation of simulation and graphics, also allows collaboration between multiple sites.

library allows for use in all other styles of VR visual displays, so it can be used at remote sites with different available hardware. Collaboration between sites, and between the visualization simulation and the renderer were both handled by the *CAVEComm* library developed at Argonne.

The *BoilerMaker* design and implementation team consisted of the principal investigators from Argonne, Lori Freitag and Paul Plassmann, and Nalco Fuel Tech William Michels, plus a team of programmers who contributed significantly to the design and implementation of the software. The programming team included Darin Diachin, Daniel Heath, Jim Herzog, and Bob Ryan, all working at the Argonne National Laboratory.

Conclusion

Initially, the *BoilerMaker* application was tested as a proof-of-concept, using public domain computation algorithms. The *BoilerMaker* application demonstrates how virtual reality can be used as an interactive design tool. By allowing

**FIGURE 3-30**

A splash screen in the BoilerMaker application gives credit to its many developers.

a designer to instantly see the results of the placement of injectors, the VR experience can aid in saving time, money, and potentially lives. *Boilermaker* also is an example of an early use of a virtual reality environment for computational steering and collaboration. The end result of this application

development effort was worthwhile. The prototype system demonstrates that the time required to analyze injector configurations for pollution control systems can be significantly reduced. The ramification of this is that better, environmentally safer boilers can be designed at lower cost.

CASE STUDY 3.5: CUTTY SARK

Executive Summary

Who could resist the lure of donning a head-mounted display, grabbing an actual ship's wheel, feeling the rocking of the ocean beneath their feet, and heading out on a high-sea adventure? Of course, such a virtual reality experience has a grand appeal, and Cutty Sark chose to create such a scenario for people throughout the United States.

Their motivations, however, were more than the altruistic idea of providing compelling entertainment. Cutty Sark

chose to develop the first virtual reality application with content specifically tailored to their product for the purpose of advertising. They spent a large fraction of their advertising budget to create a touring VR platform with an actual ship's wheel, and other accoutrements such as a handheld gun and moving deck. Computer graphics were driven by an SGI graphics engine. The plan was to allow participants to experience smuggling Cutty Sark scotch whiskey across the sea while battling pirates, weather, and other hazards. Their goal was to create an indelible brand imprint in the

minds of the tens of thousands of people who got to try the system throughout the United States. The results included increased brand recognition and increased sales.

Introduction

The *Cutty Sark Virtual Voyage* was a virtual reality experience used as part of a product promotion campaign. Designed to help promote Hiram Walker's new slogan for Cutty Sark—"The Real McCoy"—the *Virtual Voyage* puts the spectator aboard the ship of smuggler William McCoy during the days of prohibition.

Of course, as is part of any product marketing scheme, the ultimate goal is to increase sales. With generally limited means of advertising available (print media, store advertising/promotions, and billboards), a novel medium was selected to help boost awareness of the scotch—virtual

reality. After taking the promotion on the road for more than 18 months, sales that had been sliding for over a decade turned around, and brand awareness was up.

The slogan "The Real McCoy" raises the comparison of Hiram Walker's scotch vs. all other brands. Legend has it that "The Real McCoy" refers to the scotch smuggled by William McCoy during Prohibition, when lesser-quality homebrewed "whiskey" was the norm. Telling such a story in print advertising is difficult. Although it is an interesting tale, it is unlikely that many readers will expend the effort to read it. Thus Hiram Walker sought to do something different; so when Scaros & Casselman proposed a virtual reality experience, Hiram Walker was interested. Scaros & Casselman then put together a team experienced in producing VR experiences, including Horizon Entertainment and GreyStone Technology Inc.



FIGURE 3-31

Virtual reality was used as part of an advertising campaign to increase brand awareness of Cutty Sark scotch.

Application Description

The VR promotion was packaged as a pair of touring VR platforms, each equipped to run a game experience based on the legend of Captain McCoy. The storyline of the game is of the participant taking the role of a captain trying to get “his” shipment of 1000 cases of Cutty Sark scotch delivered safely. Facing numerous hazards along the journey, the experience scores participants based on how many cases they successfully transport to the destination.

The experience presents McCoy’s journey from the Bahamas to Long Island via a series of vignettes. The journey begins in a “sheltered harbor,” where novice ship pilots can get their bearings and experiment a little with the controls. They are then given 3 minutes to face many hazards—“fierce sea storms,” “marauding raider boats,” “attacking planes,” and “treacherous narrows”—before arriving at the “safe cove” with a “torch-lit welcoming party.”

Representation of the Virtual World

Presented as realistically as practical, the world of *Virtual Voyage* is a cartoon-like polygonal world on a foundation of fairly realistic physics. Objects in the world include the ship itself, its cargo, the ocean and shore, and the computer-controlled agents that serve to impede players during their journey.

During the experience, the participant is positioned at the helm of the ship, enabling them to view the world from the vantage point of the captain. In particular, they see the fore of the ship and can look out over the ocean as they navigate the world. They are also able to see an avatar of their hand (equipped with an avatar of a pistol). However, the focal visual element of this experience is the cargo of Cutty Sark scotch cases on the deck. Players keep a watchful eye on the precious cargo before them, as the loss of each case reduces their score.

In keeping with the attempt for reasonable realism, the rendered visuals are the type comfortably produced by a typical high-end graphics engine—a large number of texture-mapped polygons. This type of rendering approaches, but is not quite “photorealistic.” It tends to result in the cartoonish countenance mentioned above. Fog was used in some circumstances to allow polygon culling in scenes with high polygon counts.

Somewhat realistic audio cues (marker sounds) are heard when events occur during the adventure, helping to make the world seem more alive. Sounds were played from basic sound sample files and processed into spatialized sounds through a Beachtron™ system. By having the sounds of the boat and airplane follow the objects through the world, participants get a greater impression of being in the world of the experience.

DESIGN CHOICE

No active feedback to the other senses was provided in this experience.

Because the platform units had to be set-up and dismantled many times during their travel, the designers felt that the added technology required for other senses would increase the cost, as well as the susceptibility to failure. At the same time, it would not heighten player (mental) immersion significantly.

However, the platform structure was constructed with springs underneath the helm where the participant would stand. This was done so as participants shifted their weight in correspondence with the rocking boat, they would feel the deck moving beneath them. The developers were uncertain of the effectiveness of this method as a means of increasing realism and immersion, so provision was made to lock down the platform in the event it proved ineffective or caused motion discomfort. David Polinchock of the CyberEvent Group reports that the springs were not locked down for most people, and that he feels that the added realism of standing on an unstable platform helps immerse the player.

There are some other interesting haptic possibilities that might have increased the realism, had the technology been available. One additional sensation could have been to use the ship’s wheel give feedback indicating the difficulty of turning the rudder against the sea. Other interesting haptic possibilities include providing wind and water spray, helping to put the player in the middle of the raging storm—though of course this would require extra precautions to keep the water from coming in contact with any of the electronics.

The rolling, choppy seas could result in active vestibular feedback. A simple olfactory display might also have been an interesting addition to this application. An ambient sea-smell would require no connection to the computational system, yet would reinforce the setting in the player's mind. Also, whiffs of gunpowder smell could be generated relatively easily each time players fire their weapons.

Interaction with the Virtual World

Although the use of the *Cutty Sark Virtual Voyage* experience was for product promotion, it falls within the genre of the *mission game*. In this case, it is a game that happens to be designed with an eye for promoting a certain product. The overall form of interaction is that of *pilot-through travel*, with agent communication done through the business end of the pistol.

As mentioned, the narrative of this experience is reenacting the voyage of Captain William McCoy as you smuggle scotch from Bermuda into the U.S. The “plot” of the story runs along the lines of a 1920s smuggler who must maneuver the ship while thwarting numerous hazards in an (your) effort to get the scotch through. As the ship's captain, you are responsible for safely transporting as many cases as you can. The score indicates how successful you were in the mission.

To keep throughput high, the game was limited to a 3-minute session for each player. The game's narrative is divided into five story elements, each of which lasted for a prescribed amount of time. An introductory sequence welcomed the participant and began the story of Captain McCoy while allowing the player to get a feel for controlling the ship. They were then presented with three sequences in which they face hazards along the journey typical of those that the real McCoy may have had to endure. After bearing through the hazards, and likely losing some but not all of the cargo, the experience is brought to a close by sailing into a safe cove where they are welcomed by a party of revelers at the Gatsbys'. Sequences are segued with a fade to a still image of the product, displayed along with the player's current score.

During the main body of the game, three of four possible hazards are presented. The possible hazards are:

- “Fierce sea storms”
- “Marauding raider boats”
- “Attacking planes”
- “Treacherous narrows”

During the “fierce sea storms,” high winds and rough seas force the captain to pilot the ship to keep it from listing. As the ship rocks and rolls, the cargo slides across the bow, and if the motion is severe enough, cases will fall overboard. Lightning is also a concern during the storm, as it may strike and destroy cases of cargo. The player could reduce the chance of lightning strikes by orienting the ship such that the sail remained over the cargo—this is a faux maneuver, added to give the player an opportunity for protecting the cargo in this game experience.

Likewise during the “treacherous narrows,” the primary task is maneuvering to avoid grounding the ship, causing it to list and dump cargo. The narrows were also obscured by fog. The fog served two purposes. It made navigation through the narrows more treacherous, and it also allowed the models of the rocky seas to be more complex because polygons hidden by the fog can be culled.

The “attacking planes” and “marauding raider boats” hazards fly and sail around the player's ship, harassing and shooting at it. The player loses cargo when it is shot by the attacking entities (or accidentally by the player). The biplanes and cigarette boats can be thwarted by hitting them with a certain number of shots. After a couple of good hits, the enemy boat will burst into flame, or the plane will crash into the sea, creating a large water plume and sinking after a few moments of bobbing in the sea.

A fifth hazard is ever present. This hazard is Jocko the stowaway, who attempts to make off with his own supply of scotch.

Finally, after surviving the hazards, the user sails to the Long Island shore for the clandestine meeting at which the “Real McCoy” is delivered to the awaiting party.

The participant pilots the ship while standing on a physical platform that looks like the helm of a real sailing ship. A ship's wheel mounted on the platform is a physical prop that provides the means by which the player can control the virtual rudder on the ship. The sails automatically properly align themselves to jibe with the wind. The effects of the wind and ocean on the sails and hull of the ship determine the speed and tack of the ship.

In addition to the vehicle platform, the participant is able to control one other object—a pistol. In either hand, the user



FIGURE 3-32

Participants in the Cutty Sark Virtual Voyage game must steer their ship through “fierce sea storms” while attempting to deliver their cargo.

Photo courtesy of GreyStone Technology Inc.



FIGURE 3-33

While traversing the “treacherous narrows” the participant must avoid running the ship aground.

Photo courtesy GreyStone Technology, Inc.



FIGURE 3-34

The participant must protect the cargo by shooting at attacking planes (left) and marauding raider boats (right).



FIGURE 3-35

The experience culminates in the delivery of the cargo in the stealth of night.

can hold a tracked pistol grip fitted with a trigger under the index finger. Firing the pistol is the only mechanism the user has for nontravel interaction. However, since all of the agents the participant can interact with intend to harm them, it comes in handy. The weapon provides unlimited ammunition and does not require reloading. It is aimed by the participant physically positioning their hand, and fired by squeezing the trigger.

A significant portion of the interaction within the *Virtual Voyage* is in response to the computer-controlled agents that strive to reduce the amount of the cargo. The boat and airplane hazards are both agents that navigate around and take shots at your ship when possible. Interacting with these agents involves steering the ship to avoid their fire, and firing your own weapon to keep them at bay.

The more interesting agent is Jocko the stowaway. Jocko stows away in the hold, coming out of hiding to steal cases of cargo during any of the hazard phases of the journey. Jocko is programmed to pay attention to when the participant is involved with activities that take their attention away from monitoring the cargo—such as fighting the marauders, or just gazing out over the horizon. When the Captain (being played by the participant) is not paying attention, Jocko will make for the cargo and begin to steal it. The user may notice Jocko absconding with a case out of the corner of their eye, or when they turn their head rapidly toward the cargo. Once spotted, the participant can shoot at Jocko with the gun. Jocko cannot be killed, but if a bullet comes too close, he will return to his hiding spot.

The *Virtual Voyage* attraction travels with only one platform going to any particular venue, so was designed strictly as a one-person experience. However, the touring attraction does include a monitor to encourage kibitzing by people waiting to play and other onlookers. Thus the experience becomes a communal activity in the conversations that arise comparing scores and interesting scenarios.

As a game, there are no rigorous requirements to emulate real-world physics such as might be required by a ship-training application. However, to make the experience as engaging (and thus memorable) as they could, the designers strove for realism wherever possible. The physics of the

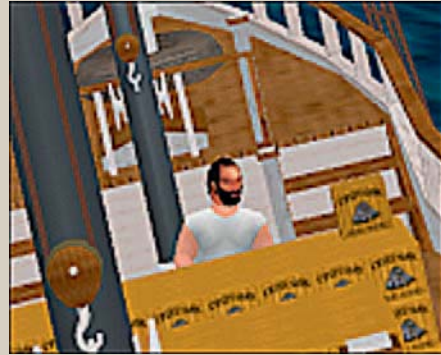


FIGURE 3-36

Jocko the stowaway lurks on the ship trying to steal the participant's precious cargo.

world simulation were accurate enough that the Captain could sail the ship using typical sailing techniques. With only a one-person crew, the sails were automatically trimmed to match the tack of the ship and the wind and sea conditions. The interface between the hull of the ship and the ocean waves and wind were also fairly realistic, producing an accurate feel of buoyancy and stability. The boat would heel drastically when inappropriate sailing technique was employed.

As the player sails through difficult conditions, the ship's cargo would slide across the bow from side to side in accordance with the rocking of the boat. When the boat was sufficiently listing, some cargo would fall off, reducing the player's score.

Collision detection, which is often very computationally expensive, is not needed to any great extent. Most of the user interactions are with physical objects that limit their motions, allowing the physical world to handle collisions appropriately. The one type of object that did require collision monitoring were bullets and determining what object was struck.

Venue

Venues included stops at restaurants, nightclubs, trade shows, food festivals, and other high-traffic (adult) locations, beginning with the National Restaurant Association



FIGURE 3-37

The Cutty Sark application uses a kiosk platform to enable participants to feel the ship's wheel and provide a general ambiance for the experience.

Trade Show and Expo in Chicago on May 14, 1994. Over the 18-month tour, the two units were booked at over 600 events covering every major U.S. market—most of them multiple times. The majority of cancelled shows were caused by logistical rather than system problems.

The touring nature of this advertisement was the major drawback for the advertising agency. Although a company specializing in such events was hired to manage the tour, bookings were typically done through liquor sales distributors. Since these distributors handle many different brands, not all of them were enthusiastic about making bookings for this single product.

Each unit handled only one player at a time. In addition to the 3-minute game time, 1 or 2 minutes were required to suit and unsuit each person. Over the course of the 18 months, approximately 55,000 people tried the system first-hand. Each player had to provide two proofs of purchase and be 21 years of age or older.

VR System

The system for the *Virtual Voyage* was designed within the constraints imposed by the cost, the tour, and the audience. In this case, there are multiple venues, and the system had to be easily transportable between sites. The primary concerns were portability and durability. The platform structure was designed to include all the hardware into a single unit. The structure included the helm at which the participant would stand, plus concealed locations to hold the computer system, etc.

SYSTEM CHOICE

A head-mounted display was selected as the visual display paradigm for this application. Specifically, they chose a relatively high-resolution HMD (640 × 480 pixels), the n-Vision Datavisor 9c.

The n-Vision Datavisor is a stereo-capable display, but the development team chose to display a monoscopic image for better rendering speed. The amount of money budgeted for the rendering system allowed only one SGI RE2 engine, and using it in stereoscopic mode would have reduced the resolution to a value below the level required by the stated system specifications. A nearby monitor was positioned to allow onlookers to follow the action.

A typical electromagnetic tracking system was selected, the Polhemus Fastrak™. Tracking was done using full six DOF positional tracking. The tracking transmitter was hidden in the ship platform to keep it from standing out obtrusively.

Two “prop” input devices were used in this experience. The most prominent prop was the platform, which was rigged to look like the helm of a tall ship, including the ship’s wheel, used to give steering input to the simulation. The other was a basic handheld pistol-grip equipped with a trigger switch and a tracking receiver used to aim and shoot at enemies in the game.

Graphics and computation were done on a Silicon Graphics desk-side Onyx with the RE2 graphics option. Sound file playback was done by the Onyx, with spatialization performed by a CRE Beachtron™ system.

A technician traveled with each unit to set up, run, power down, and do some repair work on the system. The HMD was the most fragile component of each unit, so a second HMD was kept on hand. If an HMD broke, the technician had the tools and knowledge to do some limited repair work on site. For more serious problems, the HMD was shipped back for repair.

The cost of system and software development was about \$1 million. Cost of the touring systems was \$1 million to \$1.5 million each. After adding in the cost of the tour (including promotion), the overall cost was less than \$10 million. This amount was a significant portion of the Cutty Sark advertising budget, much more than would typically be spent on a single campaign.

Application Implementation

From the time the idea was conceived to the time the system began appearing at venues, the development of the

Virtual Voyage was fairly rapid. The general notion of using virtual reality in an ad campaign for Cutty Sark came from the Scaros & Casselman advertising agency after seeing a Bubble Yum campaign that used a touring VR show in early 1993.

The Bubble Yum display did not use a virtual world created specifically about the product; it was just a device to attract crowds.

In the summer of 1993, Scaros & Casselman contacted Horizon Entertainment, which was responsible for the Bubble Yum exhibit. Horizon Entertainment suggested the possibility of creating a customized game for the Hiram Walker product that would be simple to play. Scaros & Casselman would require players to be 21 or older and provide two proofs of purchase to play. After some proof-of-concept development, Scaros & Casselman brought the idea to Hiram Walker and won the advertising account.

Once the account had been won, development on the full application began in February of 1994. Horizon Entertainment served as the project’s executive producer, bringing in other experts to help with development, including Silicon Graphics Inc., Spinnaker Design Inc., and GreyStone Technology Inc. Horizon was also responsible for the tour technical support, though the tour management was handled by the CyberEvent Group. Silicon Graphics provided the graphics/computer systems. Spinnaker, a theatrical set design company, produced the physical platform, and GreyStone provided the software and system design, development, integration, and implementation.

GreyStone Technology Inc. was specifically subcontracted to provide the VR operating system (their proprietary VR library built on SGI’s IRIS Performer), the application code (including the audio features) and the interfacing to the tracker and other input devices. This work also included designing the individual game scenarios and putting the overall experience together.

Using their proprietary VR library, GreyStone added the components necessary to model an ocean world, with a selection of interesting locations, a two-masted barque (the ship), and the physics to provide a fairly realistic sailing experience. The sailing physics were such that without the constraints of the game narrative, one could freely

sail around the world in a realistic fashion, sailing from one point of interest to another.

The ship itself was modeled by GreyStone after the brig *Pilgrim* in Dana Point harbor, based on detailed measurements and photographs of the actual sailing ship. The collected data and images were used to construct a realistic-looking virtual ship, and photographic textures were applied to the virtual model. Jocko was modeled by scanning the face of a GreyStone employee and putting it on the agent's avatar.

The original concept for the experience was simply to allow the participant to sail from one point to another in a race situation. However, the GreyStone developers pointed out that this scenario would result in the better players having a shorter, less rewarding experience. As a way of increasing interest in the product, the player should be made responsible for the safety of the product. The advertising agency liked this idea, so they pursued the experience with the more complex narrative.

During development and play-testing, players experienced each hazard sequence for 60 seconds. This provided enough time to become familiar with each new environment and learn enough skills to survive. However, as a marketing device, the number of people given the opportunity to try the experience was very important. Thus, the entire experience was limited to 3 minutes, from the time the player signaled they were ready to set sail to the display of the final score. This length meant the time spent with each hazard had to be reduced by about half, and the game was never play-balanced for the shorter times.

The shorter play sequences also meant that many of the details present in the application were not apparent to the participants. For example, the raider boat contained two men, one steering and one on the bow holding a tommy gun and balancing himself against the waves. As the boat approached, the man lowered the gun and aimed it at the player's ship. The sense of self-protection is lessened when the player is too rushed to notice such details.

Conclusion

The goal of the application was to raise product awareness, so as with any campaign, the effectiveness of the advertising was measured both in awareness levels and sales.

In follow-up surveys, 80–90% of the players were found to remember the brand name associated with the experience four months later. Overall brand awareness levels were at an all-time high, and sales that had been declining for over a decade had turned around.

It is difficult to directly measure the relationship between who participated in the experience and who makes future purchases. Gregg Zegras, account executive for the Cutty Sark campaign at Scaros & Casselman, suggests that other industries have better opportunities to make such measurements. For example, if the NHL, or specific hockey teams, sponsored a VR experience, they could compare the list of VR participants with new purchasers of season tickets.

Although it consumed a large percentage of their advertising budget, the turnaround in sales and awareness were very effective for Hiram Walker. They saw their sales increase for the first time in 15 years. In addition, the campaign itself was the winner of the Advertising Age Media and Marketing award in the "Consumer Kiosk" category.

However, the difficulty of booking a touring exhibit and moving the units from venue to venue made continuing this line of advertising impractical. Zegras feels that using VR to draw in customers can be effective in the right situations. In particular, it would be more beneficial if enough display units are constructed to allow them to remain at venues for much greater lengths of time. Again, a hockey experience located at a fixed venue in towns with professional hockey teams may be able to hit the right market, without the need for the logistical difficulties of a touring event.

Although the participants probably don't think they are actually breaking any laws by their action of smuggling "the virtual McCoy," and wouldn't have considered the dangers encountered in the experience to be life-threatening, they certainly found the experience to be memorable, as shown by the follow-up surveys. The experience was successful at accomplishing the goal of creating an application that would get people to live an experience associated with the product, and therefore remember the message (advertisement) longer.

It is vital to understand, however, that what is of the utmost importance is the story itself, not the medium. Virtual reality

may be part of why people come to expose themselves to the experience, but it is the story that draws them in and stays in their memory. Content is key. However, the participatory nature of the experience may have been compromised by the constraint of increasing throughput. Participants would be yanked from one scenario to the next without much opportunity to settle into each scene.

This was the first time a virtual reality advertising experience was created that included the product as part of the virtual world. The result was a resounding success both as

a means of advertising and in introducing the medium of VR to several thousand people. In general, Gregg Zegras believes that while the cost of using virtual reality experiences as advertising is much higher per person than traditional advertising, achieving the same impressions using direct mail would have cost a lot more. The advertisement doesn't reach 10 million people, but the thousands of people it does reach are left with an indelible impression. In this case, Zegras feels that overall the VR Cutty Sark experience "helped the brand's recognition more so than any thing else they'd done in the last 3 to 4 years prior."